EDGEFORGE 2026

FROM FINANCIAL MARKETS TO QUANTITATIVE INVESTMENT DECISIONS

Derivatives trading • multi-asset portfolios • tail risk • machine learning

MARKETS | MODELS | DECISIONS

Today’s learning journey — from markets to investment decisions

Sequence	What students will understand	Time
01 Markets	Cash and derivatives markets; pricing and execution	10 min
02 Prices & returns	Simple vs log returns; commodity curves; futures rolls	15 min
03 Trading mechanics	Long / short, margins, hedges, contract close-out	25 min
04 Portfolios	Funded assets, futures exposures and risk budget	15 min
05 Model meaning	Financial interpretation of econometric outputs	15 min
06 Downside & dependence	VaR → ES → EVT → copulas → wavelets	20 min
07 ML & governance	Trading after costs, validation, investment committee	20 min

Start with the contract and cash flow; finish with a defensible investment decision.

Your mathematical skills are the starting point — not the destination

01 An accurate forecast is useless if the instrument is not tradable at the assumed price.

02 A futures contract creates exposure, cash settlements and margin calls — not physical ownership.

03 A statistically efficient portfolio may be unsuitable if it violates a client’s loss or liquidity budget.

04 Every output must become an action: buy, sell, hedge, size, rebalance or reject.

Today: explain markets first, then interpret models in terms of cash and risk.

The competition mandate is our anchor

CLIENT

Conservative institutional investor; USD 10 million; three-year investment horizon.

INVESTABLE UNIVERSE

Global equity, bond and commodity instruments; optional currency and local exposure.

RESEARCH TASK

Choose portfolio, estimate VaR / ES / EVT, test models and stress scenarios.

FINAL OUTPUT

Reproducible Python and Streamlit application plus an investment committee memo.

The objective is a defendable investment recommendation — not a decorative dashboard.

The five questions a quantitative investor must answer

1

Market

What exactly is traded?

2

Cash flow

When do money and margin move?

3

Return

Where does profit or loss arise?

4

Risk

How bad can the downside become?

5

Decision

Is the position suitable and executable?

The instrument and its cash flows must determine the data and model.

MODULE 01

Financial markets

Understand where prices come from and how trades settle.

10 MIN

A financial market is a system for transferring capital and risk

EQUITY MARKETS

Shares transfer ownership claims; returns come from price changes and distributions.

DEBT MARKETS

Bonds create contractual cash flows; their prices respond to yields and credit risk.

FOREIGN EXCHANGE

Currencies transfer purchasing power; spot, forward and swap markets manage currency needs.

DERIVATIVES

Contracts transfer exposure to underlying prices, rates, volatility or credit events.

A trader first asks: what legal claim and economic exposure do I hold?

Cash market versus derivatives market

CASH / SPOT

Buy shares: pay cash, receive ownership.

Buy a bond: receive coupons and principal subject to credit risk.

Purchase physical commodity: storage and logistics matter.

DERIVATIVE

Buy a future: receive price exposure and daily P&L.

Enter a forward: negotiate a future exchange obligation.

Buy an option: pay premium for a conditional payoff.

The same market direction can create very different funding, liquidity and loss paths.

How a trade moves from intention to settlement

1

Quote

Bid / ask and available depth

2

Order

Market or limit order submitted

3

Execution

Price and quantity matched

4

Clearing

Counterparty / obligations managed

5

Settlement

Cash or assets move; futures mark to market

Execution price and settlement mechanism affect realized returns.

Four market participants — four different objectives

HEDGER

Producer sells futures to reduce downside exposure to commodity prices.

SPECULATOR

Takes a long or short price view and accepts risk to earn a return.

ARBITRAGEUR

Attempts to exploit price inconsistencies subject to costs, credit and limits.

INSTITUTIONAL INVESTOR

Allocates risk across asset classes to meet liabilities and return targets.

Never infer the direction of a trade without understanding why the participant holds it.

AUDIENCE CHALLENGE

One market, two motives

An airline expects to purchase aviation fuel in six months. Does it prefer a LONG or SHORT position in energy futures to hedge higher fuel prices?

DEBRIEF LONG price exposure is the directional hedge against rising fuel costs.

Hedge instruments introduce basis risk: jet fuel prices may not match crude oil futures perfectly.

MODULE 02

Financial instruments & returns

Price series are not the same as investable wealth processes.

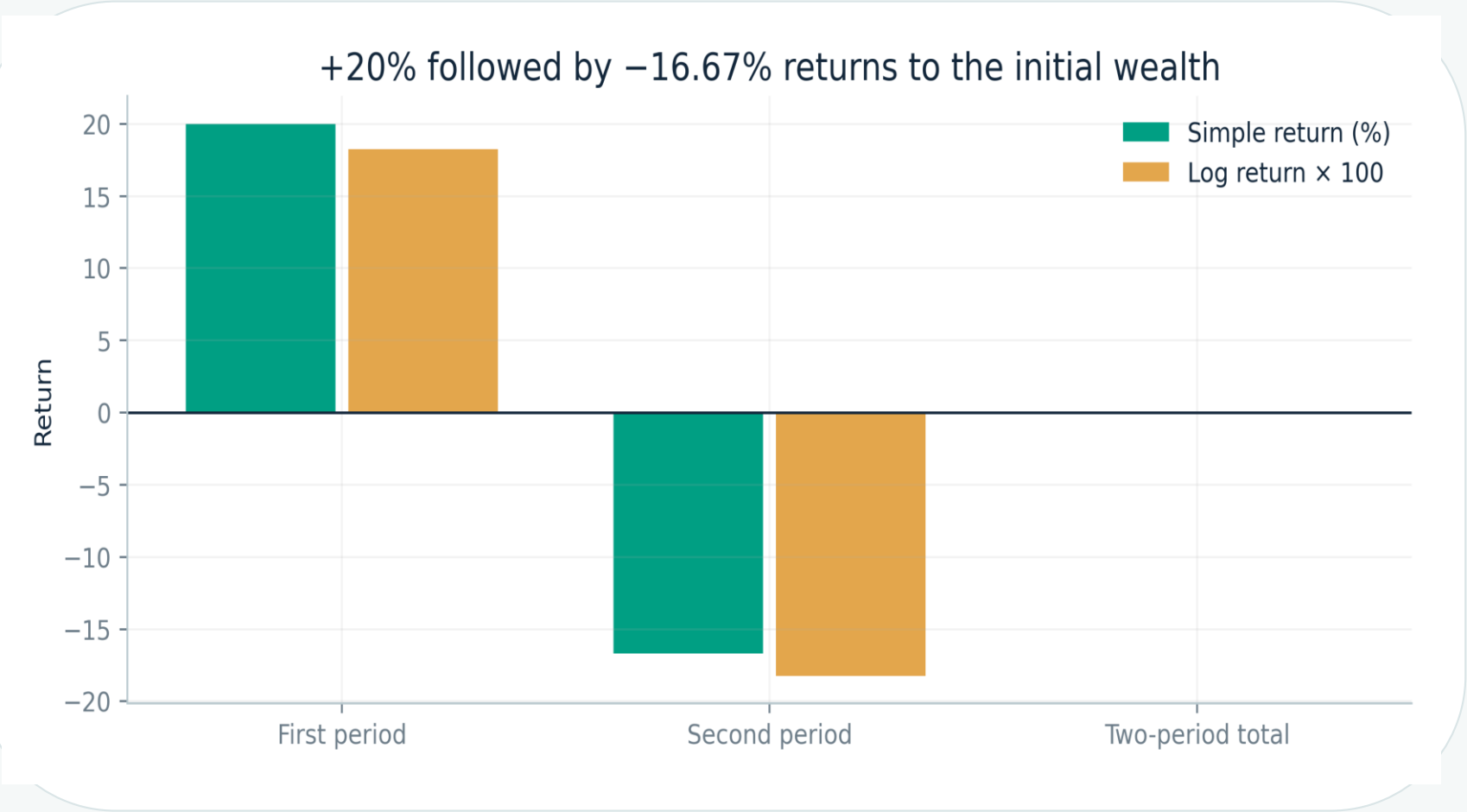
15 MIN

The return must follow the instrument

Instrument	What you observe	Economically relevant return
Equity / ETF	Adjusted prices + distributions	Total-return wealth change
Bond	Clean or dirty price + coupon	Total return; yield is not identical
Spot FX	Currency exchange rate	Change in value in investor base currency
Futures	Maturity-specific contract price	Daily Δprice × multiplier + collateral
Options	Option premium / mark	Option price change net of premium & costs

A price-only chart often omits important sources of investor P&L.

Why simple returns and log returns differ



READ THE CHART

- Simple returns compound multiplicatively.
- Log returns aggregate across time.
- Portfolio wealth and arithmetic P&L require care.

Illustrative wealth example: 100 → 120 → 100.

Which return should I calculate?

$$R = \frac{P_1 - P_0 + D}{P_0}$$

WHAT THE SYMBOLS MEAN

P_0 purchase price; P_1 sale/valuation price; D = cash distributions during holding period.

WHAT A FINANCE PROFESSIONAL DOES

Choose a total-return price series when comparing investable equities/ETFs. Adjust dividends and splits correctly.

Use the model only if its inputs match the instrument being traded.

Log returns are useful — but not automatically better

01 Time-additivity: log price changes can be summed over consecutive periods for a single positive price series.

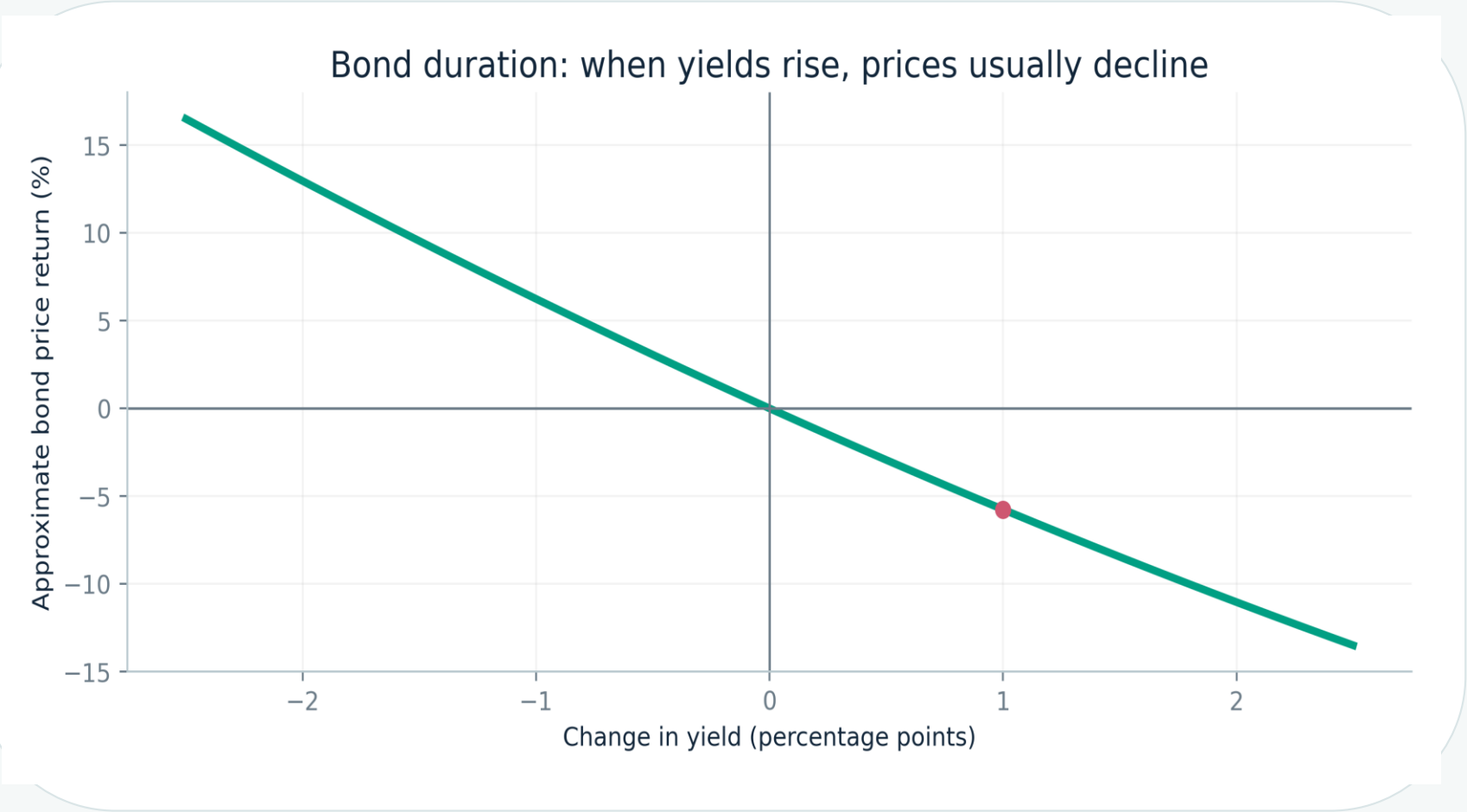
02 Statistical work: transformed positive prices sometimes improve modelling and support continuous compounding.

03 Economic reporting: investors care about compounded wealth, cash P&L and simple total returns.

04 Critical caveat: multi-asset portfolio simple returns combine by weights; futures price series may be negative.

Select the return definition based on the instrument and the decision.

Bond markets: yields and prices move in opposite directions



READ THE CHART

- Duration translates rate shocks into P&L.
- Long-duration bonds can lose sharply when yields rise.
- A “safe bond” is not necessarily a stable-price bond.

Illustrative duration model, not a live bond quotation.

Why the commodity cash price may be the wrong investment data

PHYSICAL SPOT PRICE

Quotes a quantity, grade, date and delivery location.

Trading physical oil involves transport, storage and operational constraints.

Spot indices may not reflect a scalable investor strategy.

TRADEABLE FUTURES PRICE

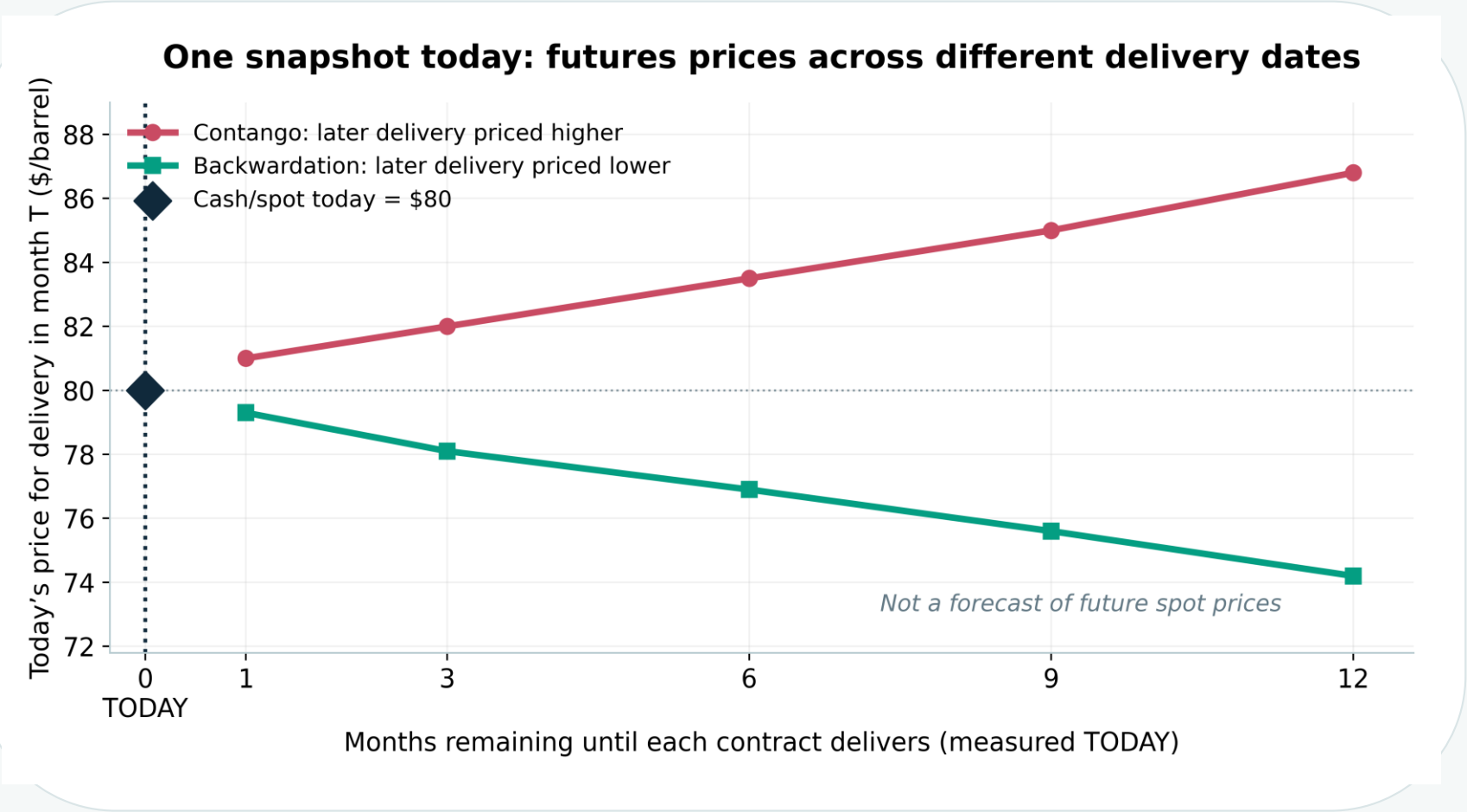
A named contract month, exchange, size and settlement rules.

Daily mark-to-market defines actual trading P&L.

Rolling introduces additional realized exposure and costs.

Use spot for physical economics; futures for a futures strategy; ETF NAV for an ETF investor.

Today’s futures curve: 0 is TODAY, not the expiry date

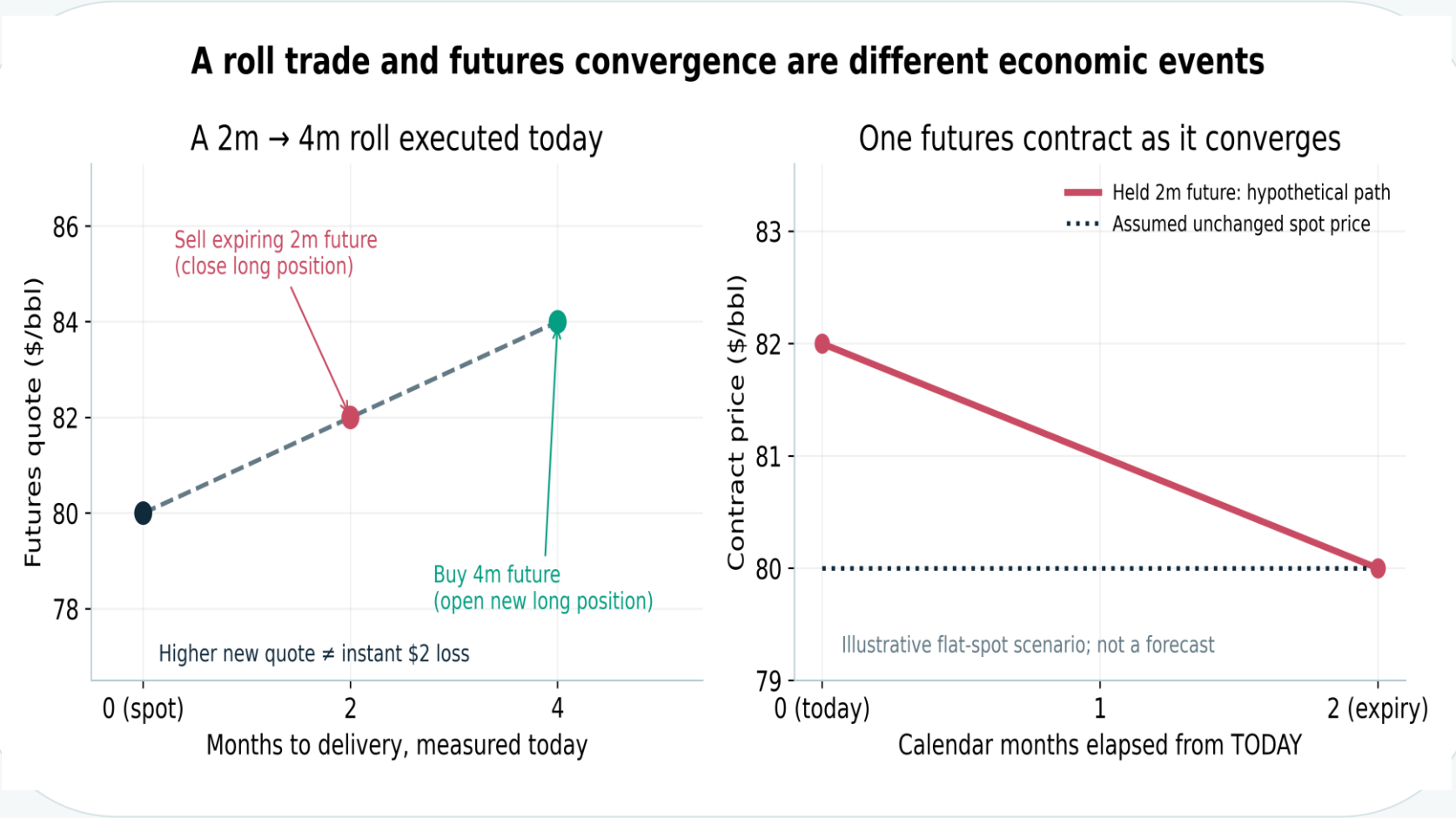


READ THE CHART

- At 0: spot price today; future contracts have T > 0.
- Contango slopes upward with maturity.
- Backwardation slopes downward with maturity.

Snapshot at t = 0: horizontal axis measures months REMAINING to each delivery date.

Rolling contracts is NOT moving along a curve in time



READ THE CHART

- Closing + opening at different quotes has no instant roll loss.
- The held futures contract gains/loses as its price changes.
- Curve dynamics, convergence, collateral and costs drive returns.

LEFT: maturity axis TODAY. RIGHT: calendar time elapsed for ONE held contract.

Two different axes: snapshot vs trading history

TERM STRUCTURE AT ONE DATE

Horizontal axis = months UNTIL expiry.

Month 0 = today (spot reference).

At 3 months: today’s quote for future delivery.

PRICE PATH OF ONE CONTRACT

Horizontal axis = calendar days/months elapsed.

At day 0: today’s entry price.

Near expiry: same contract often converges to spot.

Confusing these two x-axes creates incorrect futures trading interpretations.

AUDIENCE CHALLENGE

Does rolling a future instantly cost $2?

You are long the near future. Today you sell it at $82 and open a longer-dated future at $84. Is the $2 difference an immediate cash loss?

DEBRIEF NO. Both futures are entered/closed through variation margin and position offsets; the new contract starts near zero economic value.

Subsequent price movements, basis convergence and costs determine realized P&L.

AUDIENCE CHALLENGE

Cash price or futures price?

A team wants to backtest buying 5% notional of WTI crude oil futures for four years. Should it use WTI physical spot prices, a random front-month ticker, or a documented rolling futures strategy?

DEBRIEF Documented rolling futures returns + contract multiplier + collateral + transaction costs.

Instrument choice determines the validity of the research result.

MODULE 03

Derivatives trading in practice

Positions, counterparties, leverage, margin, hedges and roll decisions.

25 MIN

Four derivative families — different rights and obligations

FORWARD

Customized bilateral future trade; settlement and collateral terms negotiated.

FUTURES

Standard exchange-listed exposure with margin and daily settlement.

OPTIONS

Call or put rights; option buyer typically pays a premium for nonlinear payoff.

SWAPS

Exchange defined cash-flow streams; e.g., fixed versus floating interest.

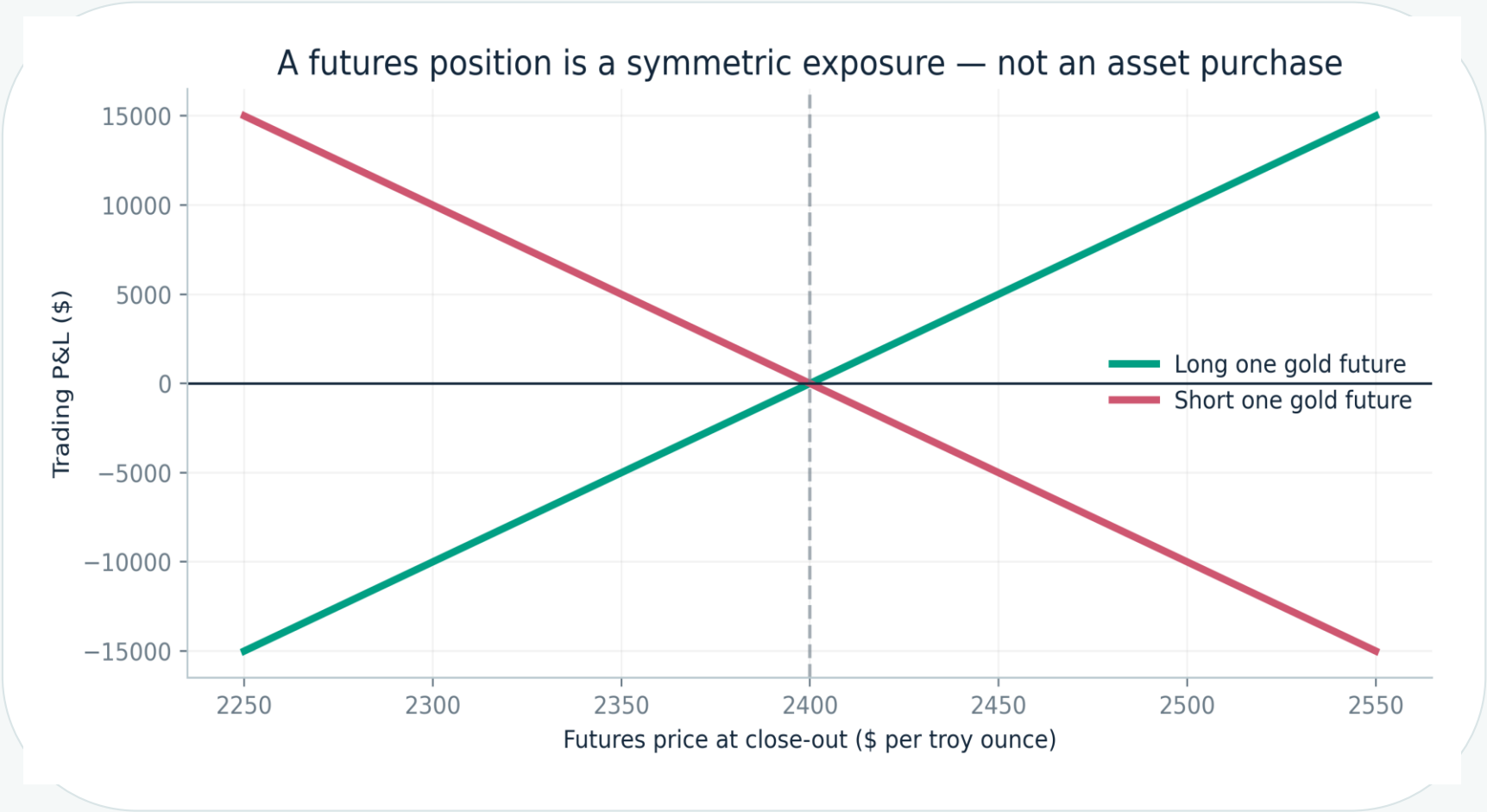
The derivative’s payoff and settlement rules determine the trader’s real risks.

Read a futures contract before trading it

Contract term	Example (illustrative)	Why it matters
Underlying	Gold / crude / EUR FX	Economic driver
Contract size	Gold: 100 oz; WTI: 1,000 bbl	P&L multiplier
Expiry	Named month / last trading day	Roll and delivery risk
Quotation	$ per oz, $ per bbl, USD per EUR	Interpret a one-tick move
Settlement	Cash-settled or deliverable	End-of-life obligation
Margin	Initial + maintenance (variable)	Liquidity and leverage constraint

A ticker alone is not a complete investment specification.

Long and short futures: symmetric daily P&L



READ THE CHART

- Long earns when the futures price rises.
- Short earns when the futures price falls.
- The loss can exceed posted initial margin.

One hypothetical 100-ounce gold futures contract entered at $2,400/oz.

Futures profit is not the movement in the margin deposit

$$P\&L = N \times M \times (F_close - F_entry)$$

WHAT THE SYMBOLS MEAN

N signed contracts (+ for long, − for short), M contract multiplier, F = futures quotation.

WHAT A FINANCE PROFESSIONAL DOES

Size contracts from portfolio risk; track realized P&L in currency and cash available for margin calls.

Use the model only if its inputs match the instrument being traded.

Futures margin versus buying shares on margin

FUTURES PERFORMANCE BOND

Trader posts initial margin; not a purchase price.

Variation margin credits/debits daily P&L.

Maintenance breaches can trigger urgent funding or liquidation.

SECURITIES MARGIN LOAN

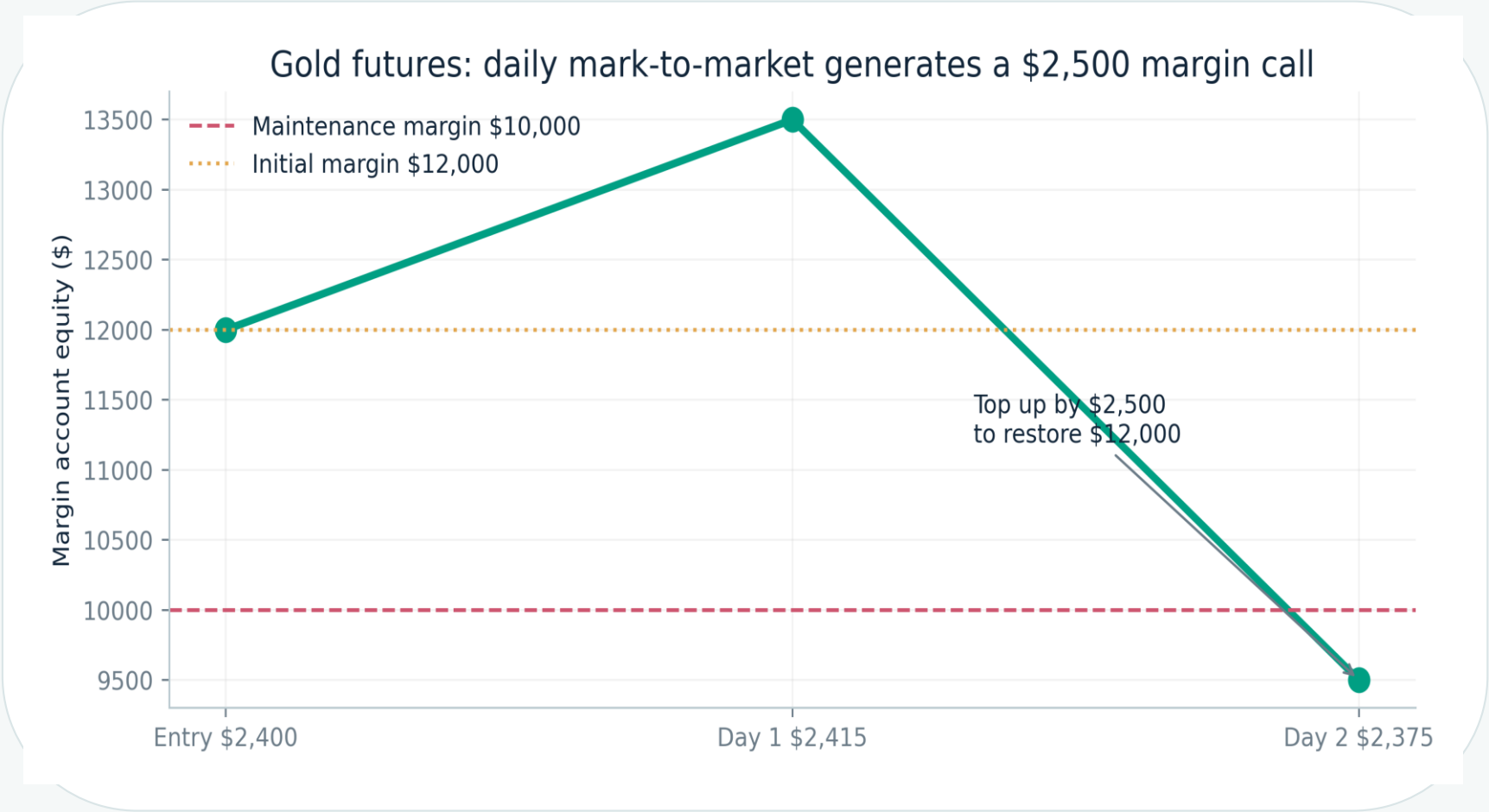
Broker lends part of purchase cost for stock.

Investor owns a financed security.

Interest and financing rules depend on the loan and broker.

Margin mechanics can dominate the timing of a profitable long-run trade.

A three-date futures margin-account walk-through



READ THE CHART

- Entry: $12,000 deposit supports $240,000 notional.
- Day 1 gain: +$1,500 → $13,500 equity.
- Day 2 loss: −$4,000 → $9,500; $2,500 call.

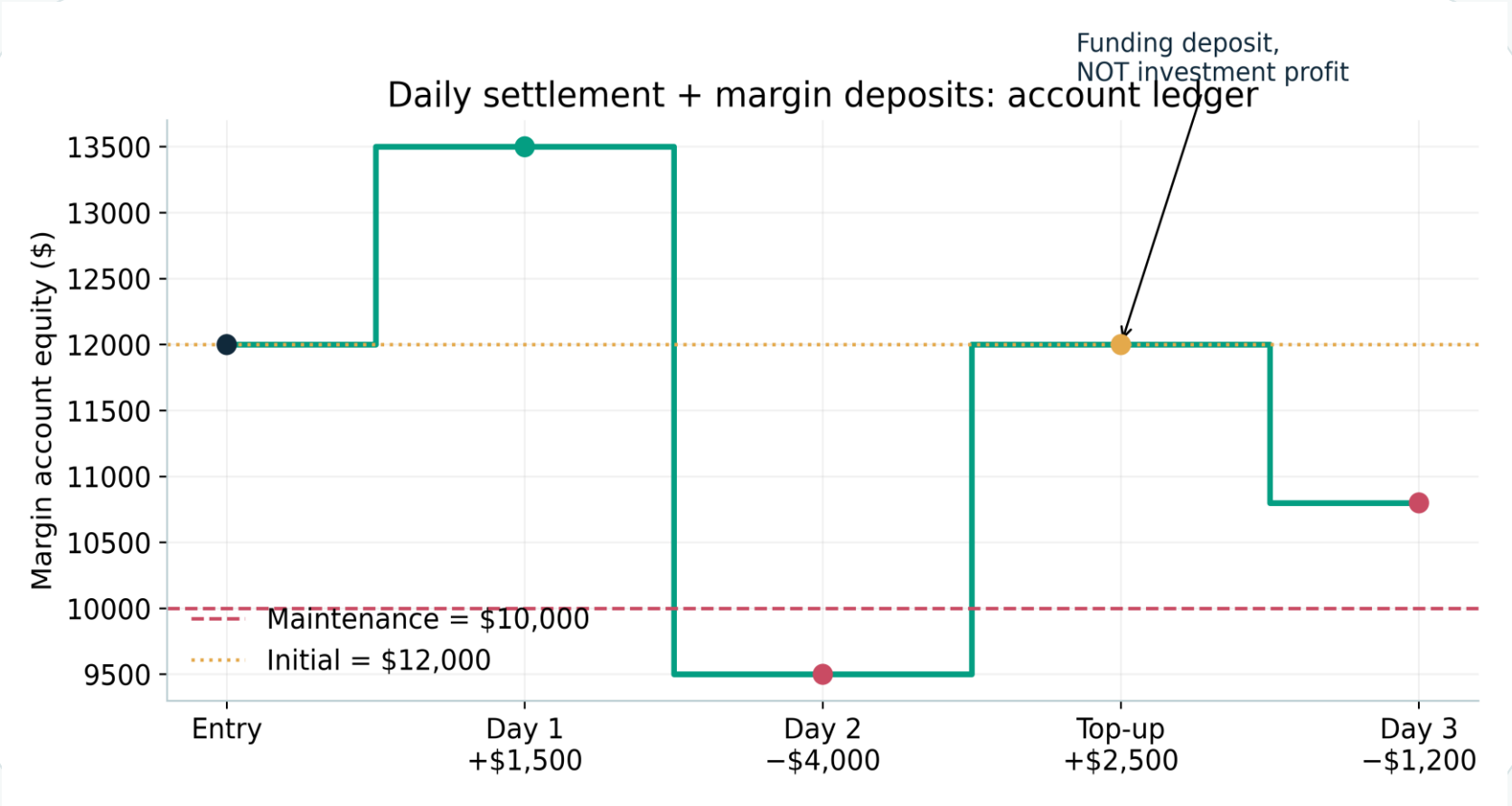
Hypothetical gold contract: initial margin $12,000; maintenance $10,000.

Margin account: numerical ledger

Date	Futures settlement	Variation P&L	Account equity	Action
Entry	$2,400/oz	—	$12,000	Post initial
Day 1	$2,415/oz	+$1,500	$13,500	No call
Day 2	$2,375/oz	−$4,000	$9,500	Below $10,000
Day 2 top-up	—	Deposit +$2,500	$12,000	Restore initial

Realized trading P&L over the two moves = −$2,500 before costs.

Collateral balance is NOT cumulative strategy profit



READ THE CHART

- Account equity includes external deposits and withdrawals.
- Top-up $2,500 is funding, not positive P&L.
- Closing exposure does not erase previously settled losses.

Hypothetical: account funds fluctuate with variation settlement AND contributions.

Short-selling shares versus SHORT futures

BORROW & SELL STOCK SHORT

Borrow existing shares through a broker.

Sell shares and face stock-borrow fees and margin requirements.

Buy back shares to cover; losses can be very large if share price rises.

SHORT FUTURES POSITION

Sell a standardized futures contract first.

Post performance bond; receive/pay daily variation margin.

Buy same expiry contract later to offset the short.

Short is an economic exposure, but funding, legal ownership and mechanics differ.

AUDIENCE CHALLENGE

A short futures hedge: demonstrate the losses too

A coffee exporter SHORTS ten commodity futures at $215 (contract multiplier 1,000 units). It closes at $225. What is futures P&L?

DEBRIEF N = −10: P&L = (−10) × 1,000 × ($225 − $215) = −$100,000.

The physical commodity value may rise, offsetting hedge losses; do not judge hedges in isolation.

How does a futures position actually end?

1

Enter

Buy 1 November oil futures contract.

2

Mark

Daily exchange settlement creates P&L.

3

Choose

Exit, roll forward, or hold toward expiry.

4

Offset

Sell 1 identical November contract to close.

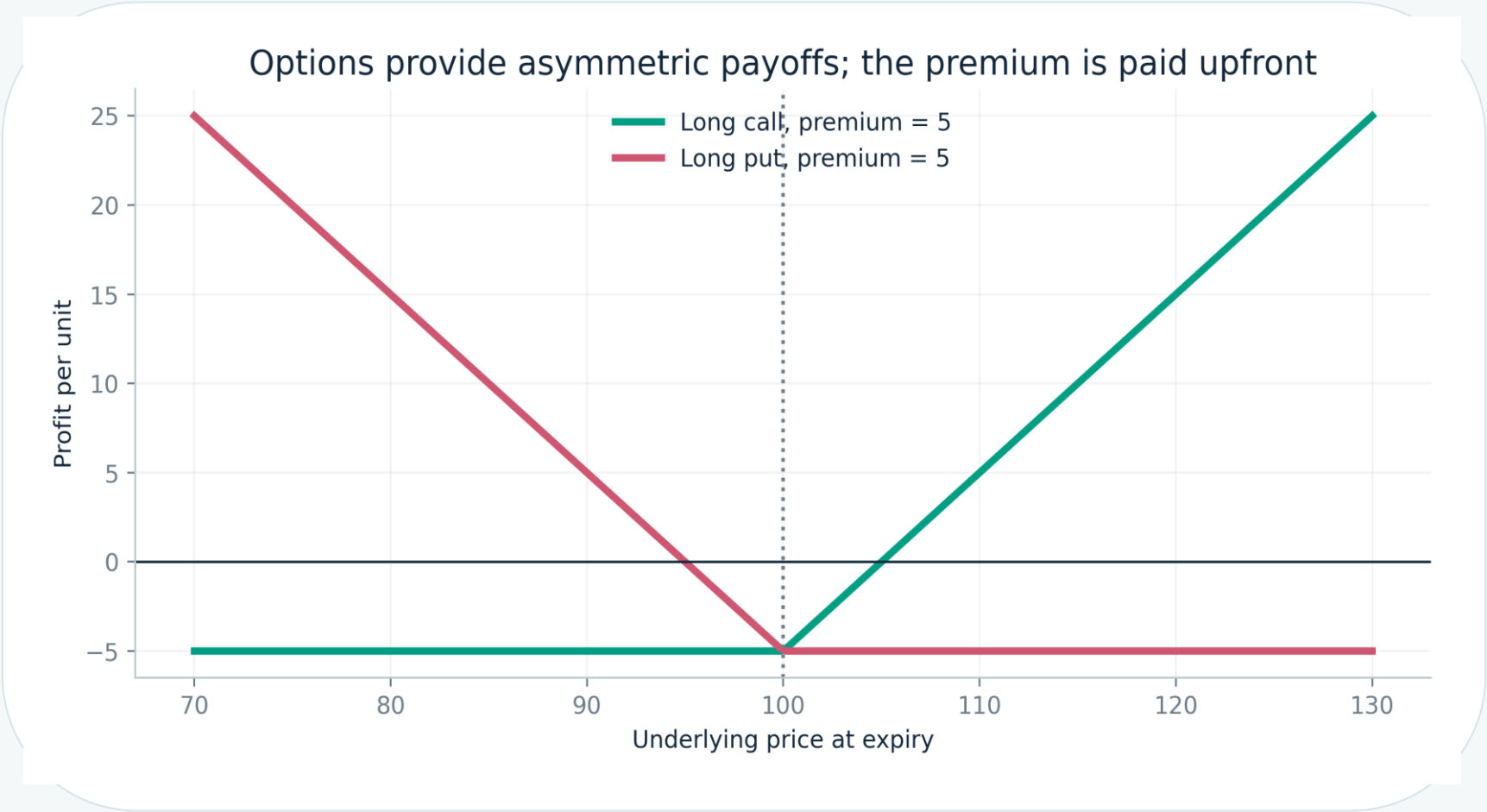
5

Check

Observe product-specific final settlement/delivery rules.

Closing a future means executing an opposite trade in the same contract.

Options are different: insurance-like convexity



READ THE CHART

- Call is the right to buy; put is the right to sell.
- Buyer downside is generally limited to premium.
- Options can hedge price gaps that futures cannot cap.

Illustrative call and put with exercise price 100 and premium 5 per unit.

A trader’s contract checklist before pressing BUY or SELL

EXPOSURE

Which economic price, market, grade, currency and contract multiplier?

ORDER

Limit versus market, tick size, exchange fees and liquid trading hours?

MARGIN

Initial/maintenance cash, stress variation calls and spare collateral?

EXIT

Offset same expiry, roll to next contract, or eligible delivery?

Contract-level mechanics determine whether the model forecast is implementable.

Commodity derivatives: who buys and sells?

OIL PRODUCER

Short oil futures to reduce revenue loss when oil prices fall.

AIRLINE / REFINER

Long relevant energy derivatives to hedge input prices rising.

PORTFOLIO MANAGER

Long or short commodity futures for tactical exposure and diversification.

RELATIVE-VALUE TRADER

Trades crack spreads, calendar spreads or cross-commodity basis.

For hedgers, gain on derivatives often offsets loss elsewhere.

A Sri Lankan importer faces a USD payable

UNHEDGED USD PAYMENT

Payable: USD 1,000,000 in six months.

If LKR depreciates, more LKR are needed.

Uncertain exchange-rate cash outflow.

OTC FX FORWARD

Agree a bank forward exchange rate today.

Lock the LKR value of the future USD purchase.

Face bank credit, forward pricing and possible settlement terms.

Do not confuse an OTC bank FX forward with a listed local exchange future.

Interest-rate derivatives: protecting financing costs

FLOATING-RATE DEBT

A company pays reference rate + spread.

Rising short-term rates increase interest expense.

Treasury wants certainty about cash debt service.

PAY-FIXED / RECEIVE-FLOAT SWAP

Floating receipts approximately offset variable borrowing exposure.

Company replaces floating risk with fixed rate payments.

Counterparty, collateral and basis risk remain.

Derivatives often change a risk profile rather than create an outright investment.

Why spot and futures prices differ

$$F(T) \approx S_0 \exp[(r + u - y)T]$$

WHAT THE SYMBOLS MEAN

S_0 spot; r financing rate; u proportional storage costs; y convenience yield; T time to maturity.

WHAT A FINANCE PROFESSIONAL DOES

Read basis and the curve before choosing contracts, setting hedge ratios or projecting roll exposure.

Use the model only if its inputs match the instrument being traded.

Choosing the market instrument — a practical investment decision

Exposure objective	Tradeable choice	Important hidden risk
Own US equities	Total-return equity ETF	Dividends, expense ratio, tracking
Own gold exposure	Physical gold ETF or GC futures	Custody / roll / margin
Tactical crude-oil view	Specific WTI futures contract	Roll, delivery, volatility
Hedge USD payable	Bank OTC FX forward	Credit, FX basis, settlement
Hedge rate exposure	Bond future / interest-rate swap	DV01, basis, contract expiry

The right backtest uses the same instrument the trading desk can execute.

AUDIENCE CHALLENGE

Close, roll or deliver?

You are long December WTI futures and want the oil exposure for January but do not want delivery. What two orders are required?

DEBRIEF SELL the December contract and BUY the next selected contract month.

A roll changes contracts, may incur spread/cost and changes curve exposure.

Currency derivatives: exporter, importer and trader

Participant	Currency risk	Possible hedge
Sri Lankan importer	USD invoice due: USD/LKR may rise	Long USD forward or futures
Sri Lankan exporter	USD receipt due: USD/LKR may fall	Short USD forward or futures
Global fund	Foreign assets create base-currency FX exposure	Hedge FX notional to target ratio
Currency speculator	Profits/losses from exchange rate changes	Long or short FX derivative

A currency hedge converts an uncertain rate into a more predictable cash-flow outcome.

MODULE 04

Multi-asset portfolio construction

Cash investment, derivative overlays, hedging and risk budgets.

15 MIN

Construct a $10m portfolio in two layers

FUNDED ASSETS

$4.5m equities + $3m Treasury ETFs + $1.5m gold ETF + $1m cash.

DERIVATIVE OVERLAYS

Illustrative $0.5m oil futures notional and $0.5m EUR futures notional.

LIQUIDITY BUFFER

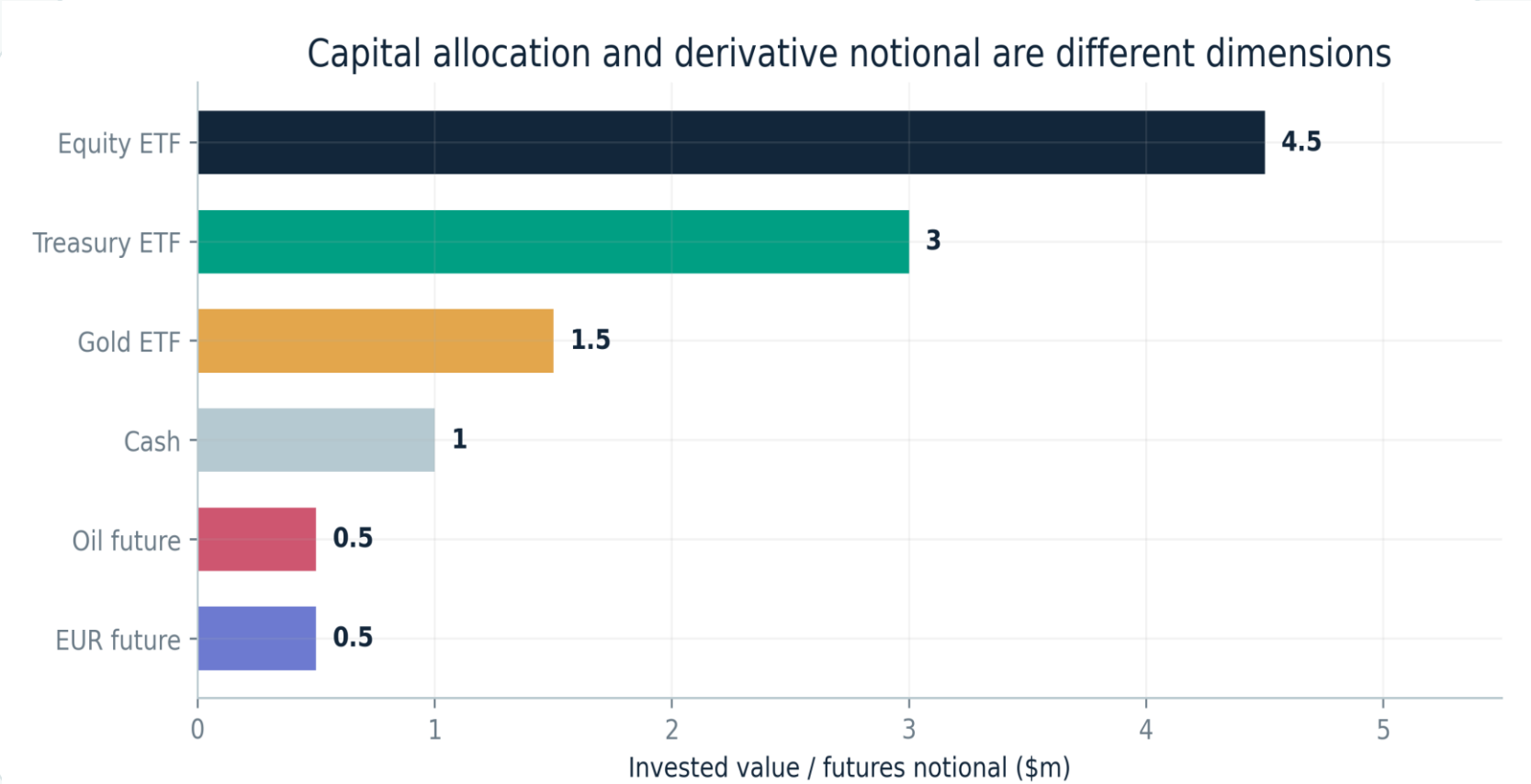
Cash pays collateral requirements, financing costs and stressed margin calls.

RISK LIMITS

Maximum drawdown, daily risk budget, concentration and stress-loss tolerance.

Derivative notional is exposure — not the dollars invested or the possible maximum loss.

How much is invested, and how much risk is overlaid?



READ THE CHART

- Funded assets = $10m.
- Derivative overlay = $1m notional.
- Gross economic exposure may exceed NAV.

Illustrative funded portfolio assets plus futures notional in USD millions.

Portfolio P&L must include derivatives correctly

$$\Delta V \approx \Sigma(q_i \, \Delta P_i) + \Sigma(N_j \, M_j \, \Delta F_j) + \text{cash yield}$$

WHAT THE SYMBOLS MEAN

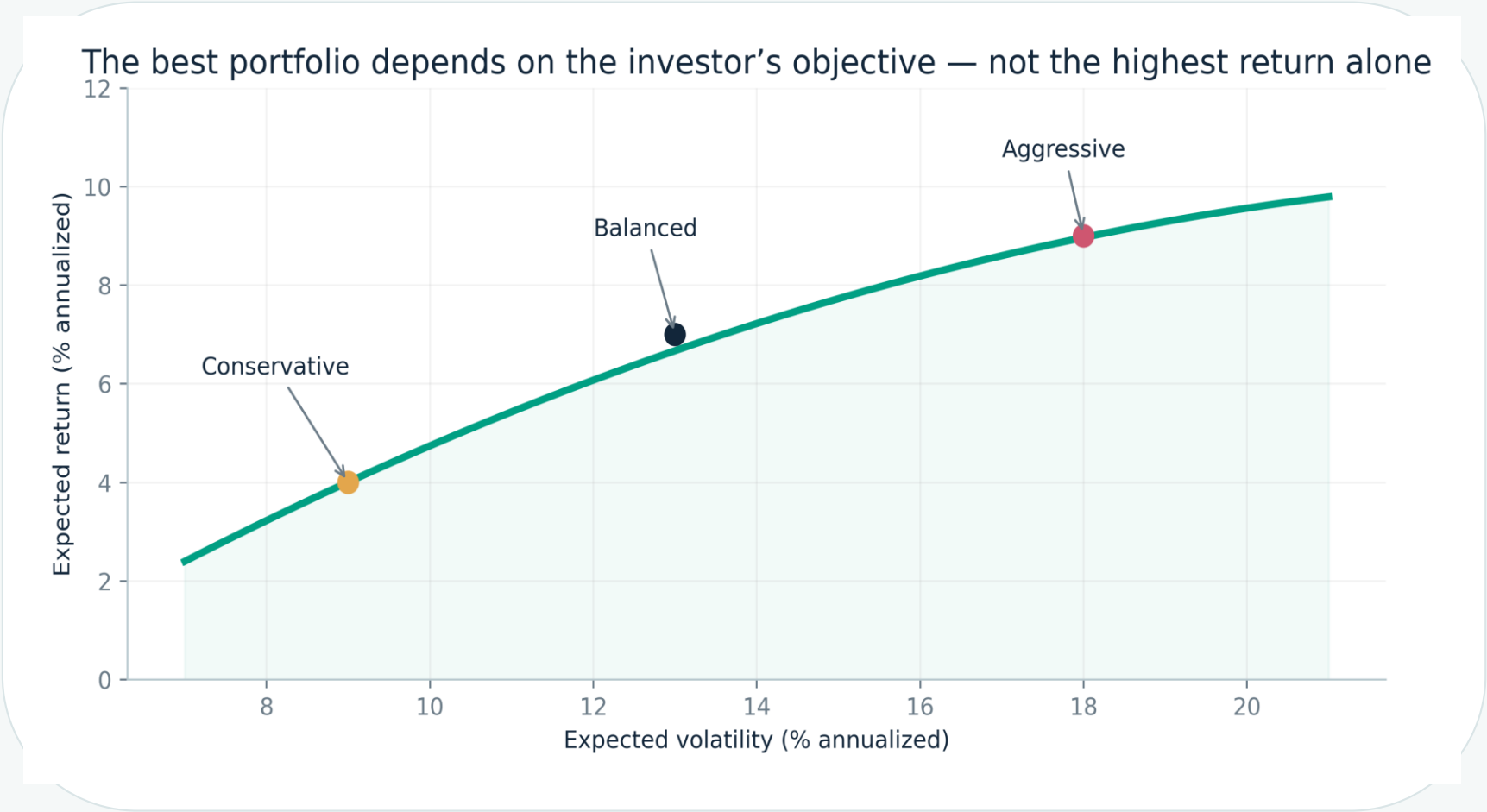
q_i securities units; N_j signed futures count; M_j contract size; P and F instrument prices.

WHAT A FINANCE PROFESSIONAL DOES

Revalue every real position, then divide total P&L by beginning portfolio NAV for the simple return.

Use the model only if its inputs match the instrument being traded.

Efficient frontier: different mandates mean different portfolios

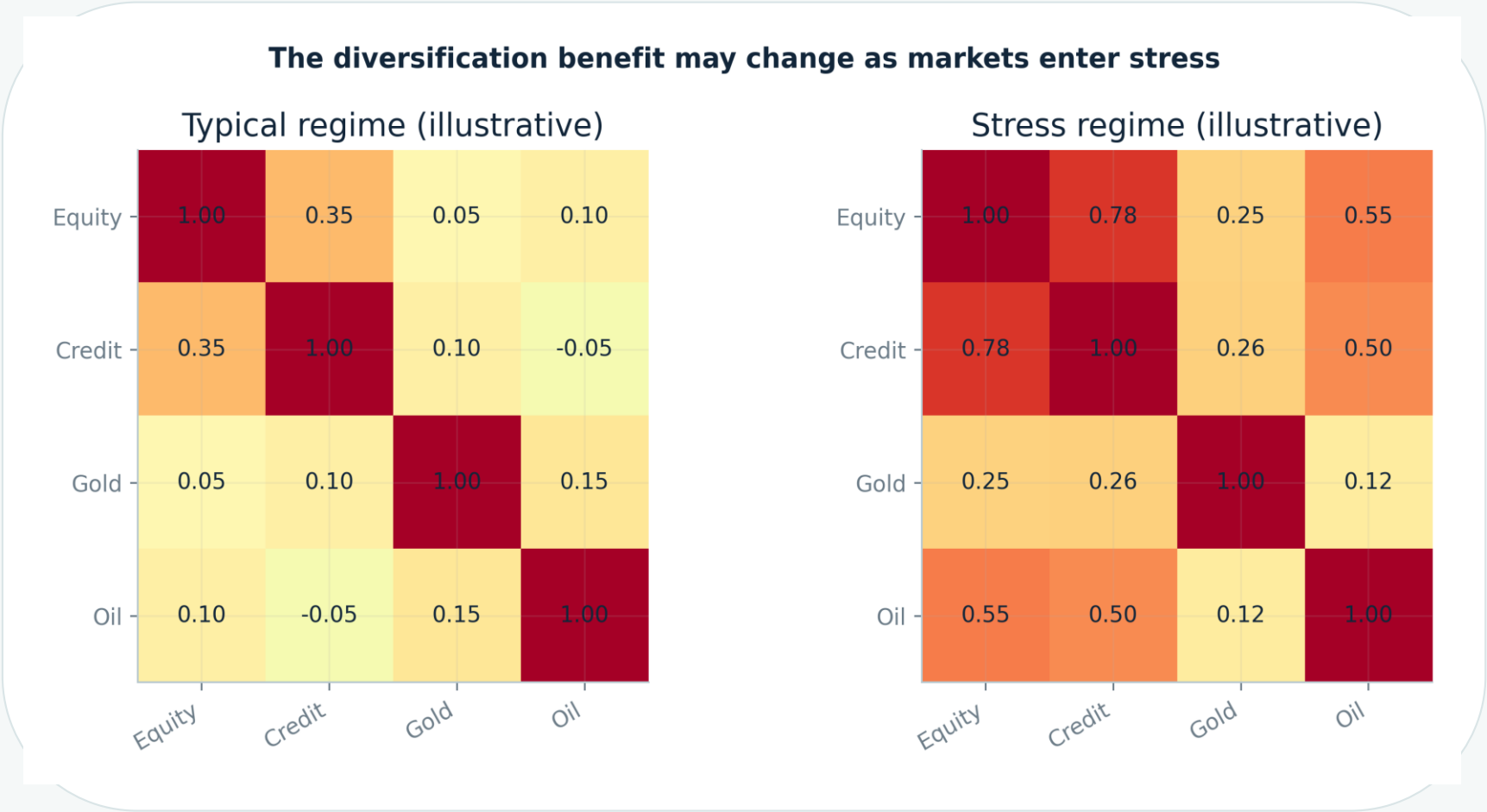


READ THE CHART

- Risk tolerance selects a feasible frontier point.
- Historical optimization is sensitive to unstable inputs.
- Include turnover, liquidity and future margin constraints.

Conceptual frontier: expected return and expected volatility, not verified market data.

Correlation may change when diversification is needed most



READ THE CHART

- Low average correlation helps diversification.
- Regime changes can create common-loss episodes.
- Stress correlation matters for multi-asset risk budgets.

Illustrative correlations; not estimates of specific real market periods.

Institutional constraints change the optimal answer

01 Concentration: per-asset and sector allocation limits, maximum futures notional and leverage.

02 Liquidity: bid–ask spreads, turnover, trading volume, margin calls and transaction costs.

03 Suitability: a conservative client may prioritize capital preservation over peak expected return.

04 Governance: scheduled rebalancing, independent model validation and explainable risk limits.

An efficient portfolio is not investable until implementation constraints are satisfied.

Notional exposure is NOT cash spent

BUYING AN ETF

Invest $1m: cash decreases $1m.

The ETF has a market value and provides total returns.

Cash dividend and securities lending effects depend on product.

LONG OIL FUTURES WITH $1m NOTIONAL

Post eligible collateral, perhaps a small fraction of notional.

Keep residual cash invested or as available liquidity.

Each $1/bbl move × contracts × multiplier affects total NAV.

Do not treat initial margin as the capital allocation or price of a futures contract.

MODULE 05

Interpreting model outputs

Translate regression, volatility and optimizer results into market language.

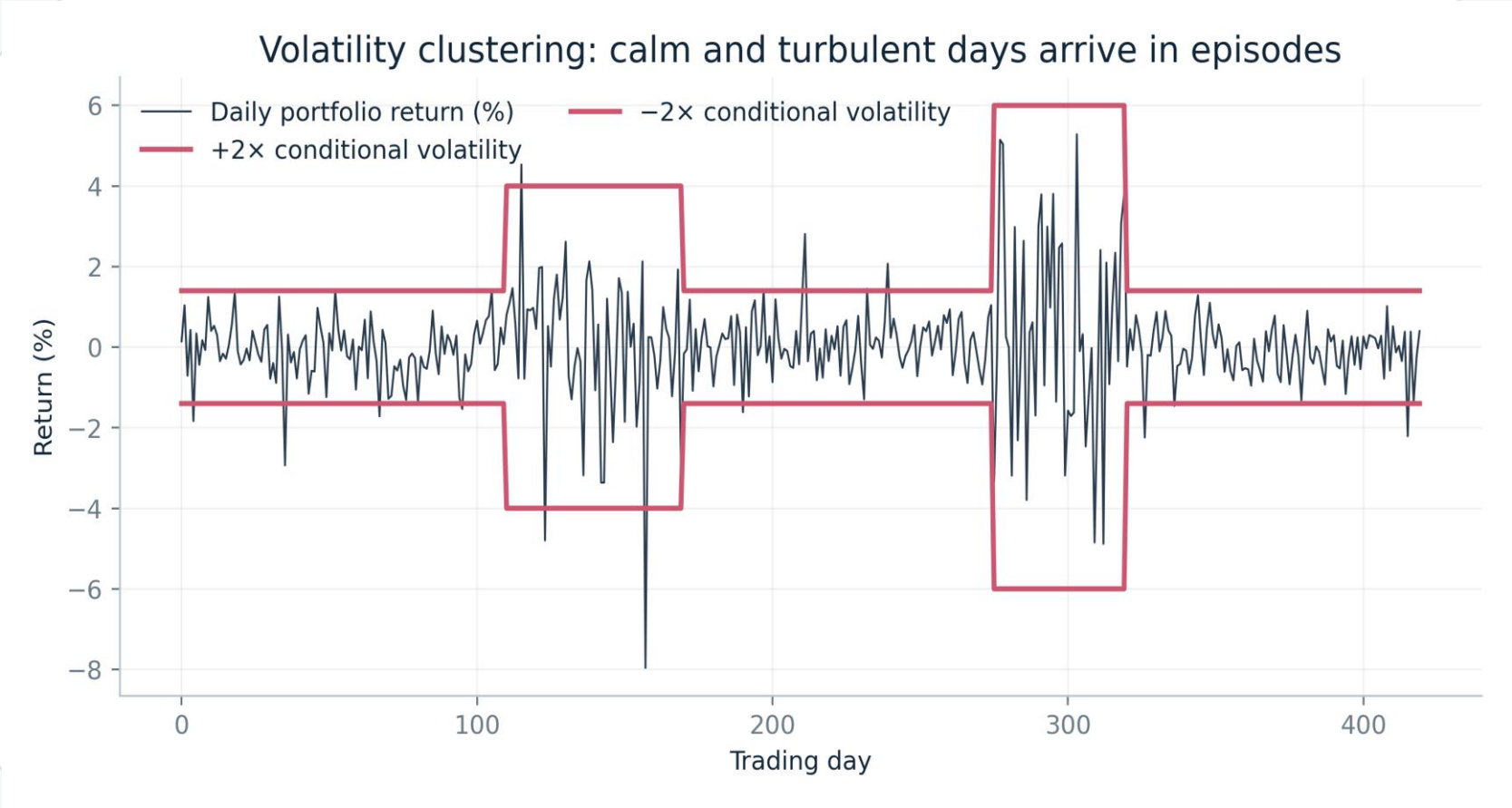
15 MIN

Read a model like a trader or portfolio manager

Output	Finance interpretation	Possible decision
Expected return μ	Estimated economic reward, not a promise	Rank strategies; size conservatively
Volatility σ	Typical dispersion of returns	Adjust leverage / risk budget
Beta β = 1.3	About 1.3× market sensitivity in model	Consider beta hedge
Correlation ρ	Average linear co-movement	Diversify, but examine tails
GARCH forecast	Time-varying conditional volatility	Reduce notional in risk spikes
Information ratio	Active return vs active risk	Compare managers/strategies

Never present a parameter without units, horizon, uncertainty and action.

Volatility clustering: what GARCH actually tells a risk desk



READ THE CHART

- Higher conditional σ means larger near-term loss scale.
- It can trigger lower exposure or tighter limits.
- It does not predict the sign of the next return.

Illustrative financial returns with calm and stressed volatility regimes.

Model forecast versus financially actionable signal

STATISTICAL OUTPUT

Predicted next-day return: +0.30%.

Forecast volatility: 2.0% daily.

Confidence intervals overlap zero.

INVESTMENT DECISION

Expected alpha must exceed commissions, spreads and slippage.

Size position using risk budget and stop-loss framework.

Decide whether estimate is stable enough to trade.

A significant coefficient need not produce a profitable or implementable strategy.

From forecast to position sizing

1

Signal

Model suggests ↑ / ↓ or expected return

2

Cost

Deduct spreads, commissions, impact

3

Risk

Evaluate VaR / ES / stress loss

4

Size

Choose contracts or units

5

Monitor

Rebalance, hedge, exit

The model cannot place a trade until risk limits and executable sizing are defined.

Interpreting econometric outputs as trades

Statistical output	What it actually says	Finance action / caution
GARCH σ = 2.5% / day	Forecast dispersion, not direction	Size down; check margin liquidity
Beta = 1.4	Equity-market sensitivity vs benchmark	A hedge may require index short
α = +0.20%/month	Expected abnormal return conditional on model	Compare all-in costs & estimation error
ρ = 0.15	Modest average linear correlation	Check joint crash exposure
ES99 = $390,000	Average loss in worst 1% model tail	Test against client risk budget

The best statistical fit does not always lead to the best investment decision.

MODULE 06

Risk beyond volatility

Why VaR, Expected Shortfall, EVT, copulas and wavelets exist.

20 MIN

Six different risk questions — six different tools

VOLATILITY

How variable are returns on an ordinary day?

VALUE-AT-RISK

What loss threshold is exceeded with a specified probability?

EXPECTED SHORTFALL

How large are losses in the worst probability tail?

EXTREME VALUE THEORY

How can we model rare tail exceedances?

COPULAS

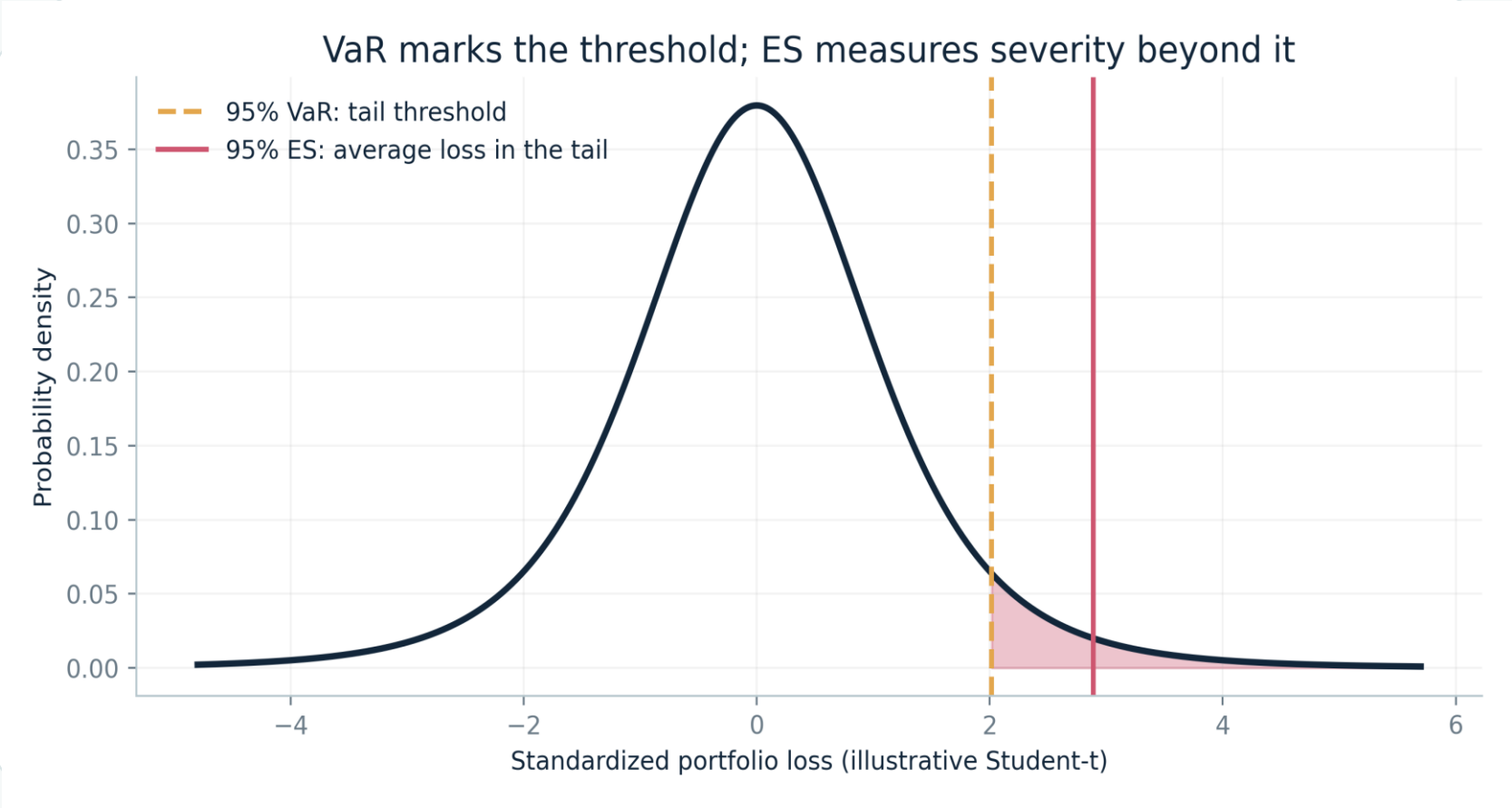
How do assets crash together beyond linear correlation?

WAVELETS

At which investment horizons and dates do dependencies change?

Each model answers a distinct decision question; none is a universal risk oracle.

VaR versus Expected Shortfall: an investor’s distinction



Illustrative fat-tailed loss distribution.

READ THE CHART

- 99% VaR = quantile of 1-day loss distribution.
- 99% ES = average of losses in the worst 1%.
- Both require horizon, sign, units and validation.

Translate a portfolio VaR into a cash decision

$$99\%\ 1-\text{day VaR} = 2.4\% \times \$10\text{m} = \$240{,}000$$

WHAT THE SYMBOLS MEAN

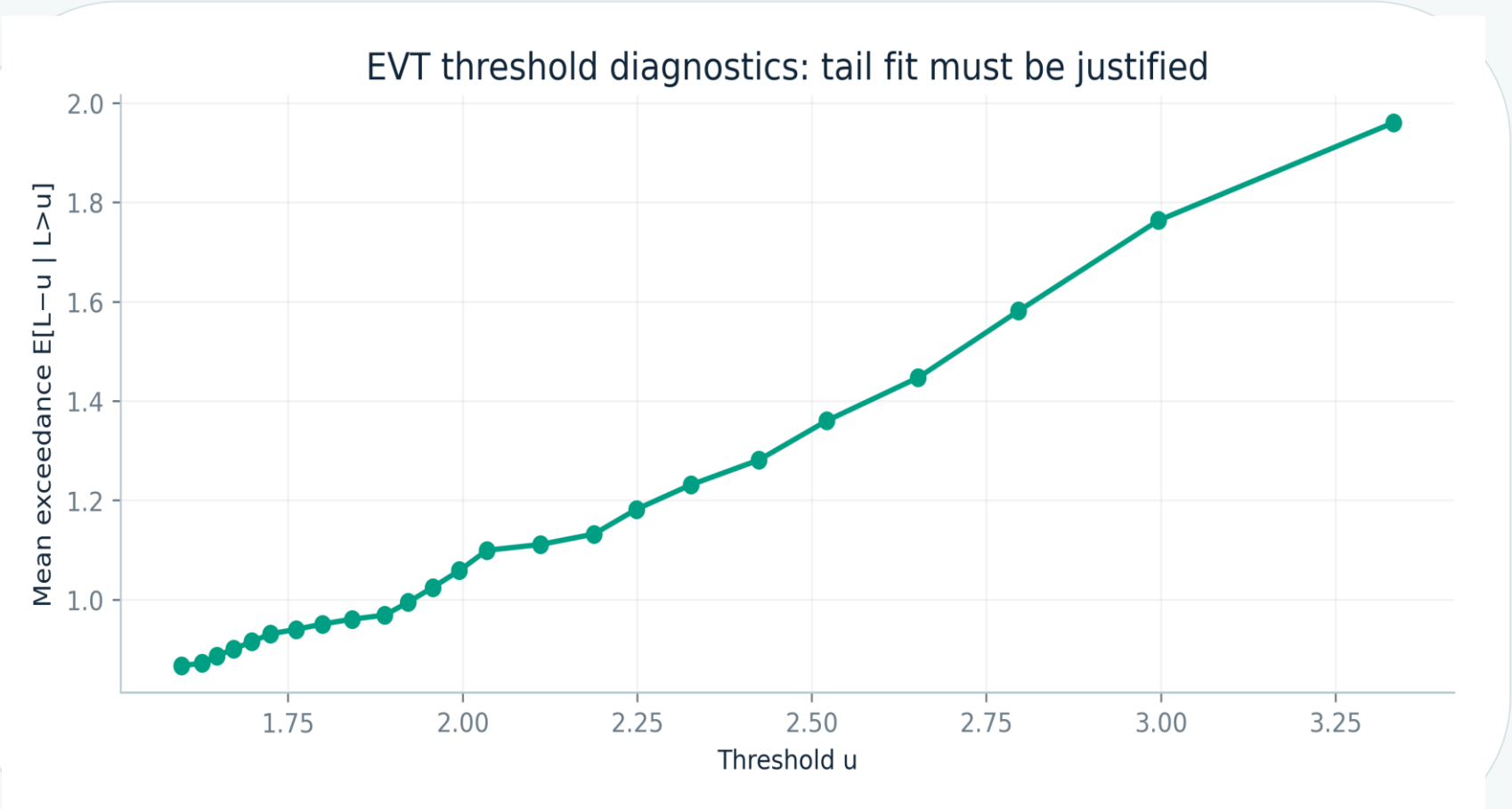
99% is the confidence level; horizon is one trading day; $10m is portfolio NAV.

WHAT A FINANCE PROFESSIONAL DOES

Check whether the investment committee can accept that loss threshold and whether margin liquidity is sufficient.

Use the model only if its inputs match the instrument being traded.

Why EVT? Normal models can misread rare-event tails

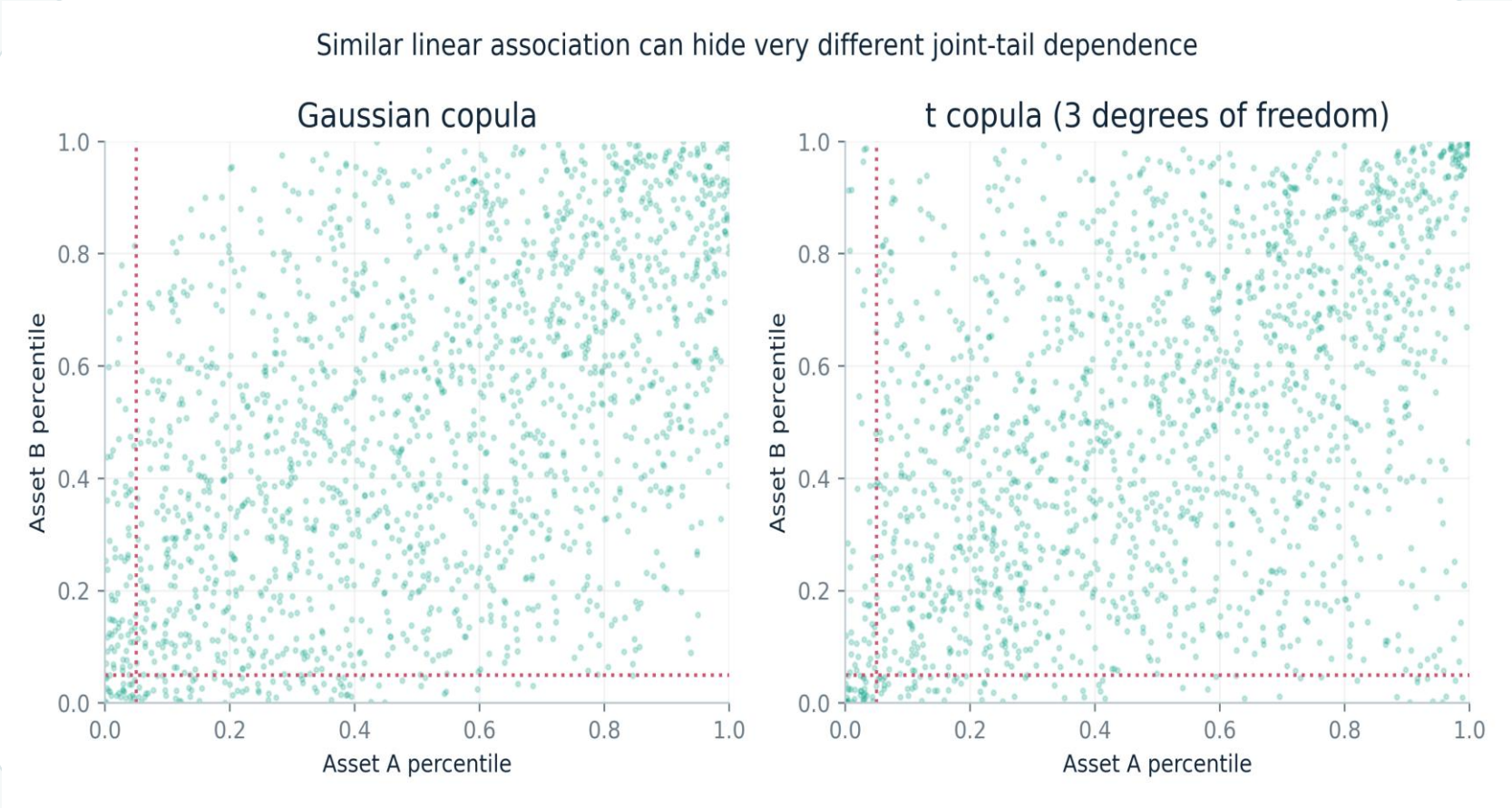


READ THE CHART

- Focus model attention on large loss exceedances.
- Estimate tail shape and scale with uncertainty.
- Choose a threshold by diagnostics, not convenience.

Stylized mean-excess diagnostic from a heavy-tailed sample.

Why copulas? Correlation does not tell the whole crash story

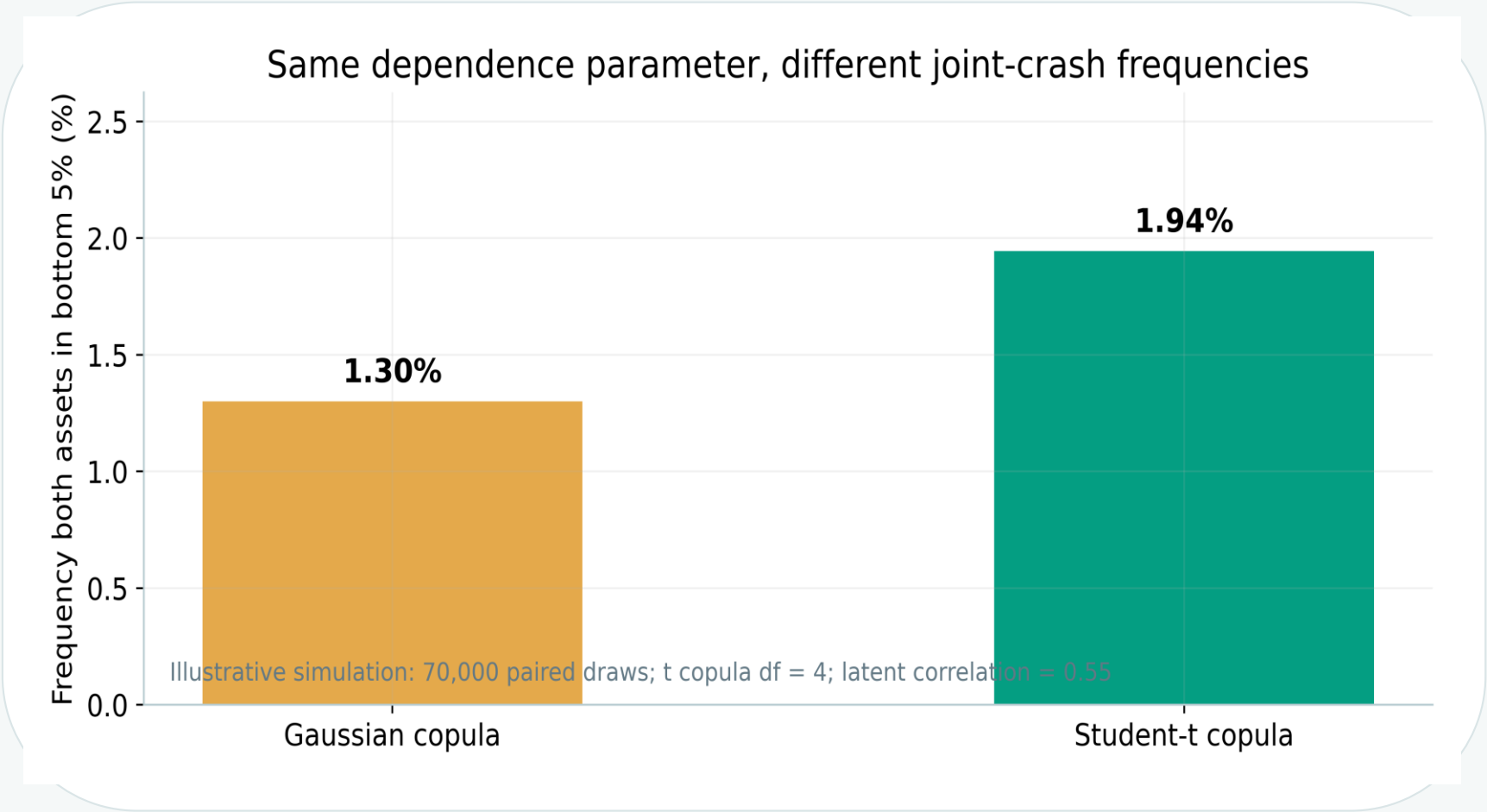


Illustrative Gaussian vs Student-t dependence on the uniform-margin scale.

READ THE CHART

- Correlation describes average linear dependence.
- Copulas describe a joint dependence structure.
- t-copulas can capture asymptotic joint tail dependence.

Same correlation parameter; different joint-crash odds



READ THE CHART

- Both dependence models can have similar central behavior.
- Student-t copula creates more simultaneous lower-tail outcomes.
- Portfolio ES and capital buffers respond to tail dependence.

Monte Carlo illustration: Gaussian vs t copula using matched latent correlation parameter.

Correlation versus copulas: what changes in a risk decision?

CORRELATION (ρ)

One scalar summary of linear co-movement.

Works well for portfolio variance given σ and covariance.

May miss nonlinear dependence and extreme joint losses.

COPULA

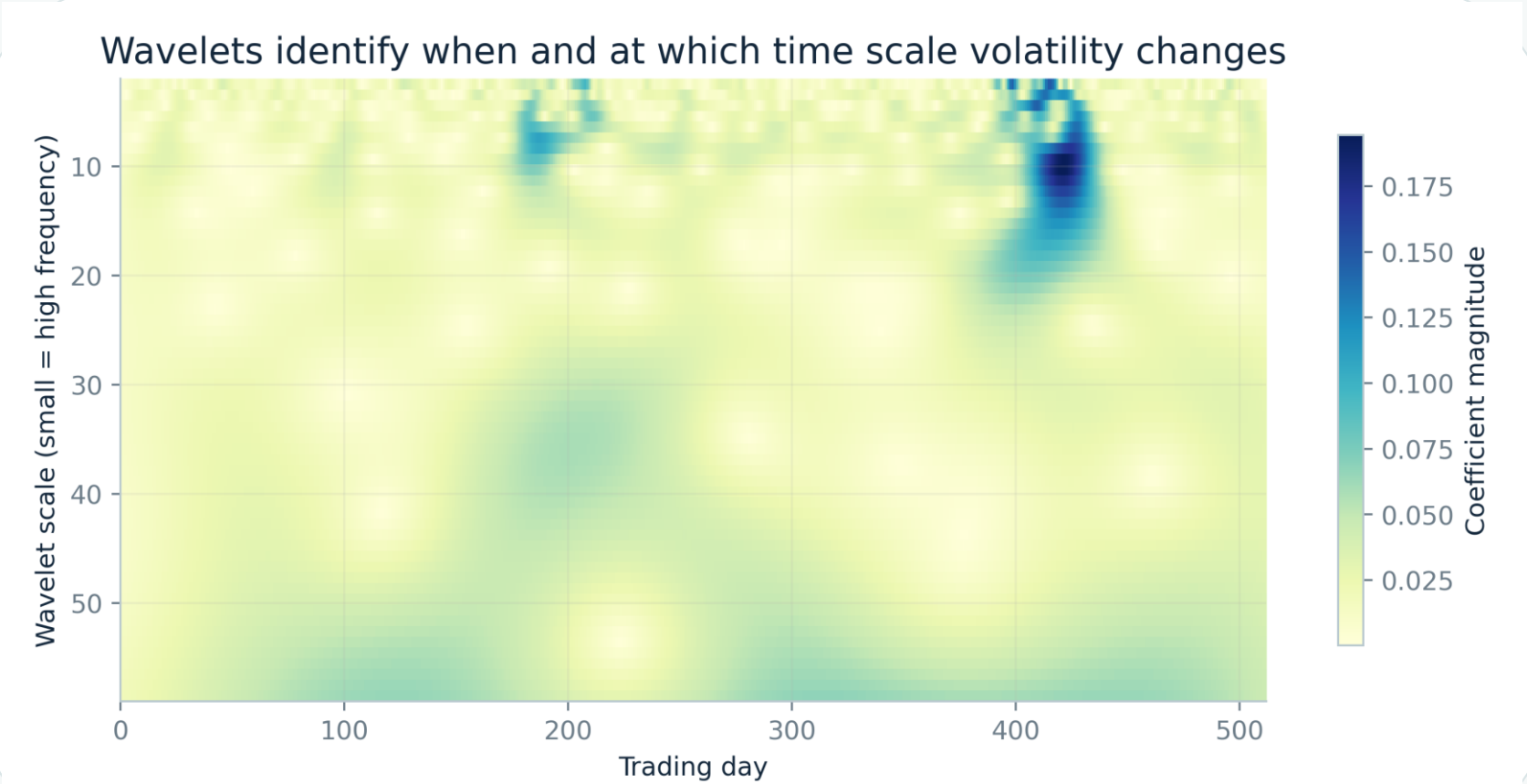
Links marginal return distributions to joint probability.

Can model joint crashes and asymmetric tail dependence.

Requires fit, goodness-of-fit, diagnostics and scenario checks.

Use covariance for normal-regime variance; use tested tail models for joint-loss stress.

Why wavelets? Risk behaves differently across horizons

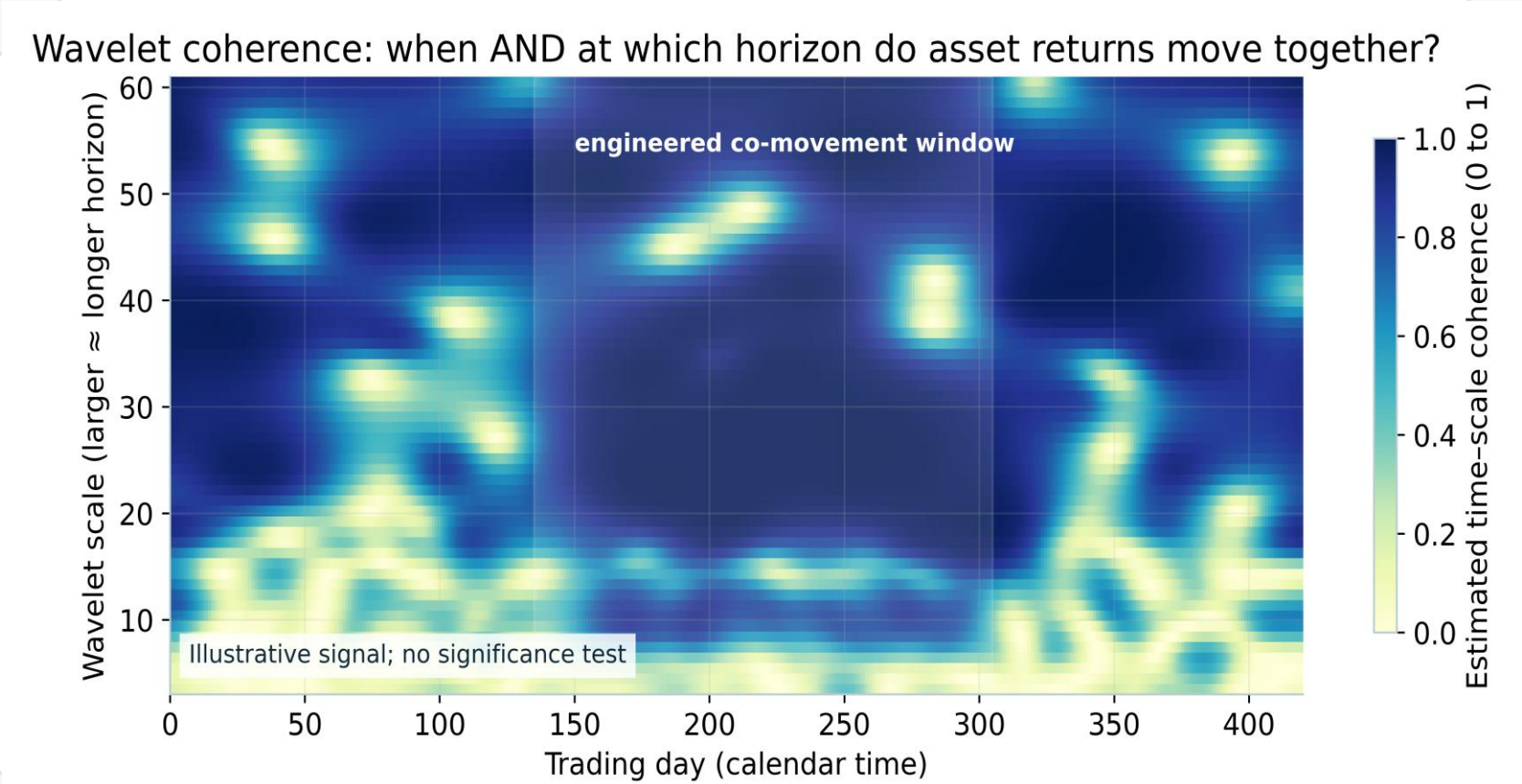


READ THE CHART

- Separate short- and long-horizon fluctuations.
- Locate when co-movement intensifies by scale.
- Useful for horizon-specific hedging research.

Illustrative time–scale decomposition of a simulated return signal.

Wavelet coherence: where joint-risk changes over time



READ THE CHART

- Horizontal axis: WHEN in calendar time.
- Vertical axis: investment scale / approximate horizon.
- Strong coherence: co-movement at that time and scale.

Illustrative two-return simulation with an engineered co-movement regime.

When would a trader choose wavelets over rolling correlation?

ROLLING CORRELATION

Simple time-windowed measure.

One average co-movement per window.

Mixes short and long investment horizons.

WAVELET COHERENCE

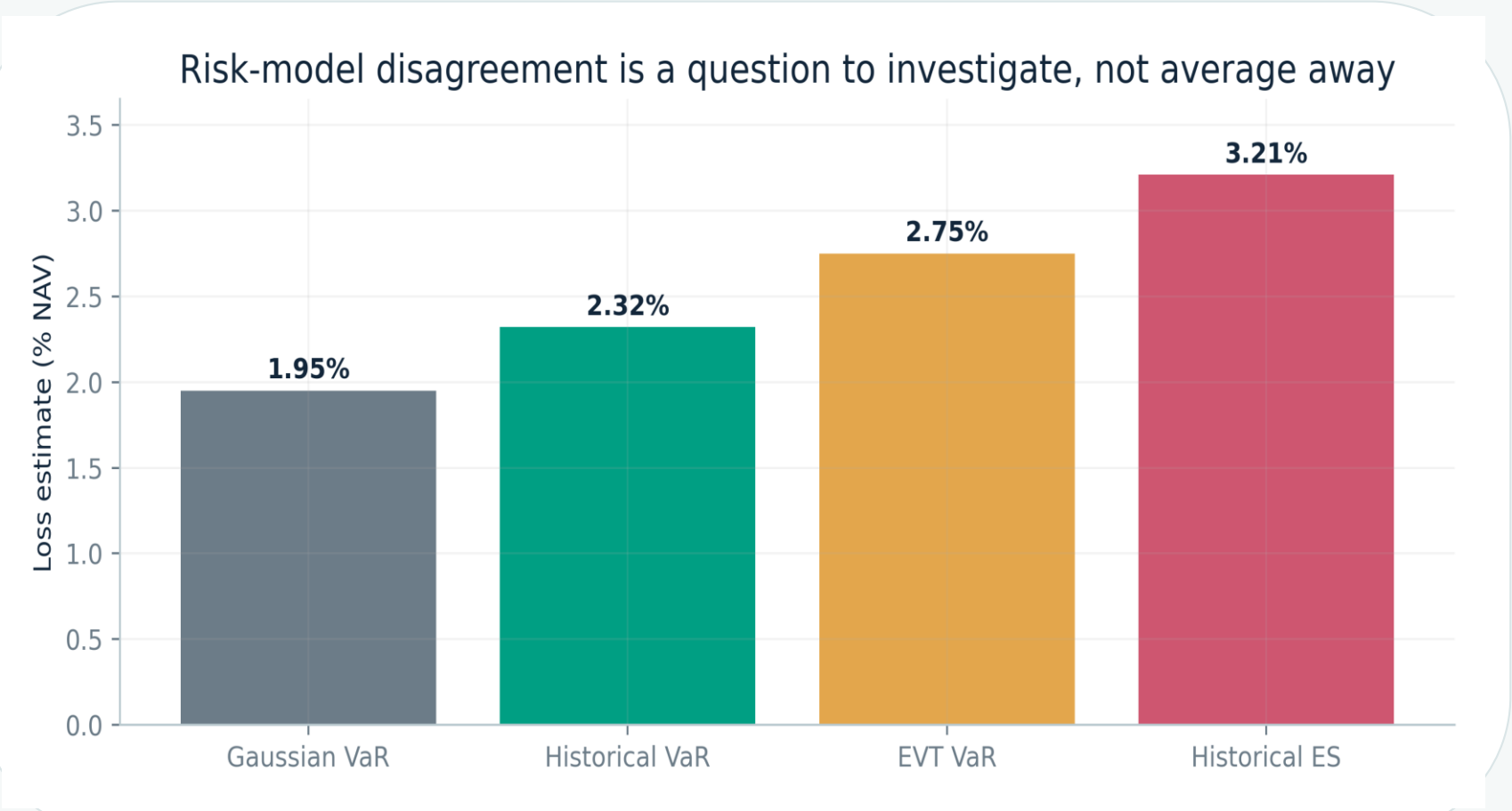
Examines time AND scale of relationship.

Can separate intraday/weekly/monthly dependency changes.

More complex, sensitive to edge and filtering choices.

Wavelets answer a horizon question; they do not automatically improve trading P&L.

When four risk models disagree, do not simply average

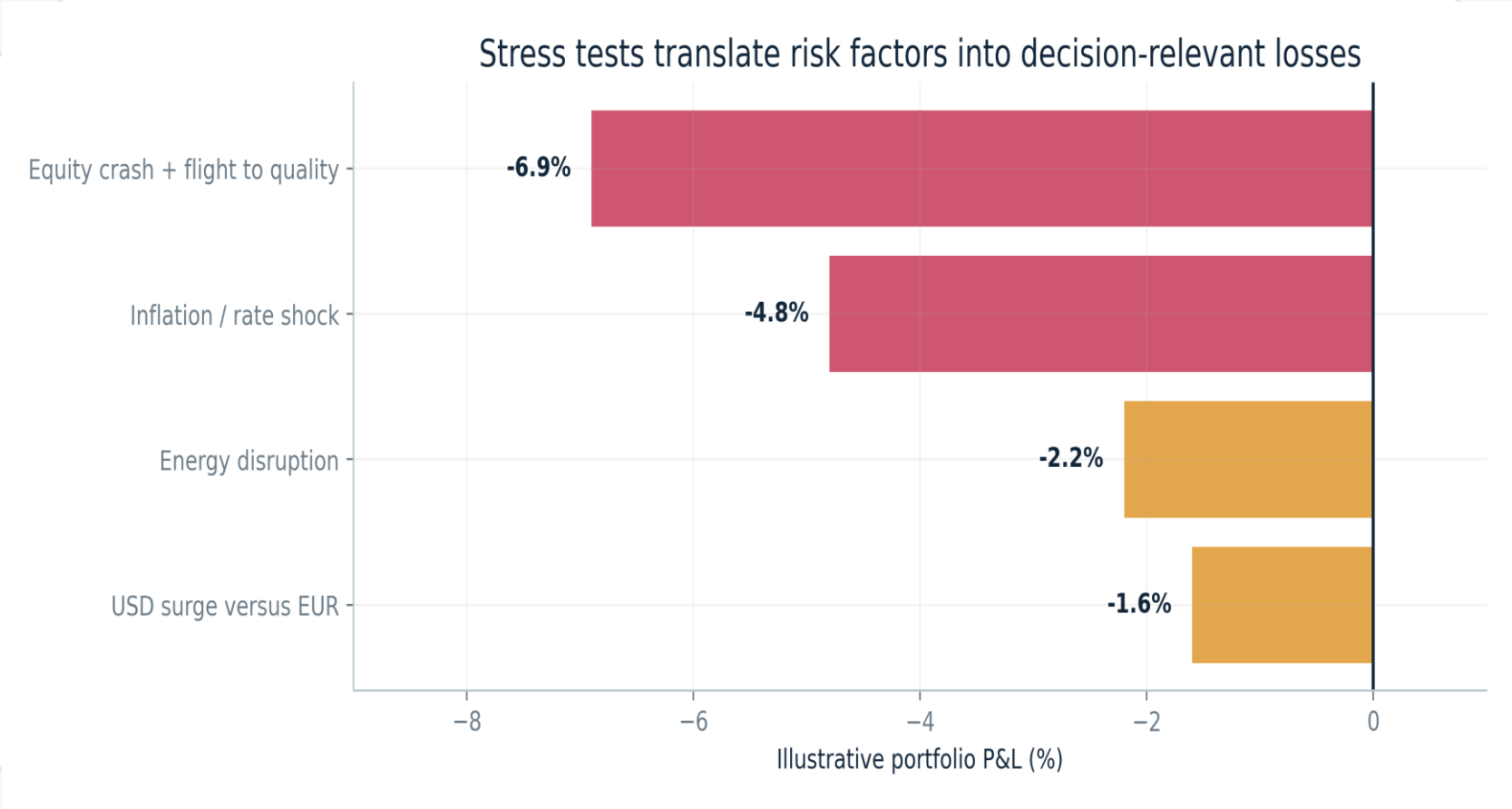


READ THE CHART

- Check units, horizon and confidence calibration.
- Investigate model assumptions and tail sample size.
- Decide whether higher risk estimates warrant de-risking.

Illustrative risk estimates, with different estimators and potentially different risk measures.

Stress tests connect model risk to board-level decisions



READ THE CHART

- Different scenarios expose different concentration risks.
- A scenario loss limit may bind before daily VaR.
- Stress losses are conditional “what if” outcomes.

Hypothetical portfolio scenario P&L, not empirically estimated crisis impacts.

MODULE 07

Machine learning & decisions

Build, test and explain a strategy that could actually trade.

20 MIN

What can ML do in quantitative finance?

RETURN SIGNALS

Predict conditional returns or next-period ranking; start with baseline.

DIRECTION / REGIME

Estimate probabilities of rising markets or risk-on / risk-off regimes.

RISK FORECASTS

Predict future volatility, correlations, default or liquidity states.

PORTFOLIO POLICIES

Transform probabilities into constrained positions, hedges and rebalance rules.

Judges should reward incremental value after trading costs, not algorithm novelty.

ML in finance — from prediction to a REAL trade

1

Problem

Which future price/return/risk is forecast?

2

Features

Use data available before each trade

3

Baseline

Compare AR / linear / GARCH method

4

Decision

Turn forecasts into positions

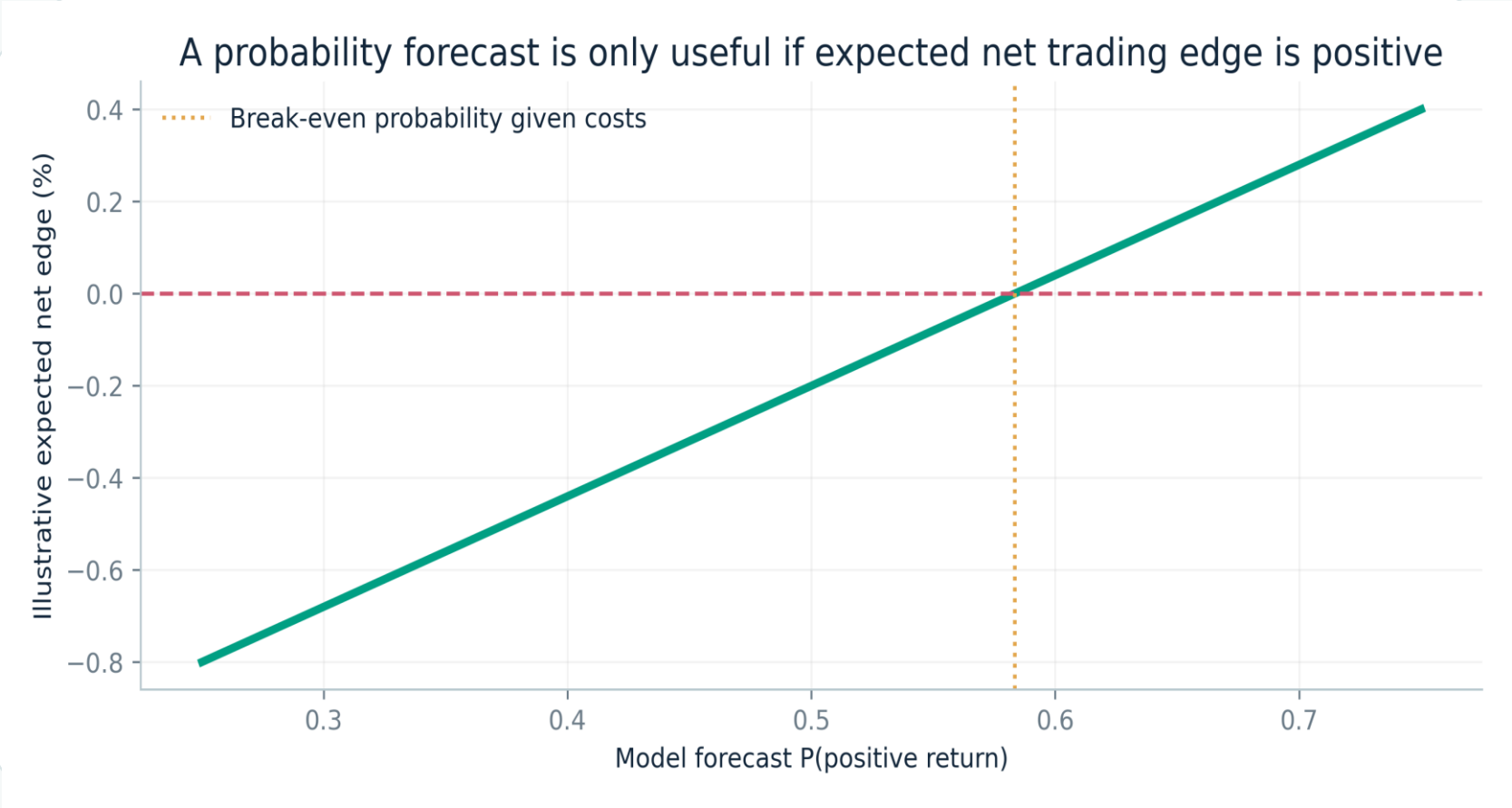
5

Backtest

Costs, rolls, margin & out-of-sample risk

Model complexity matters less than realistic economic value added.

ML probability is not equal to expected trading profit



READ THE CHART

- Forecast probability must be calibrated.
- Expected payoff includes win size and loss size.
- Subtract brokerage, spreads, financing and impact.

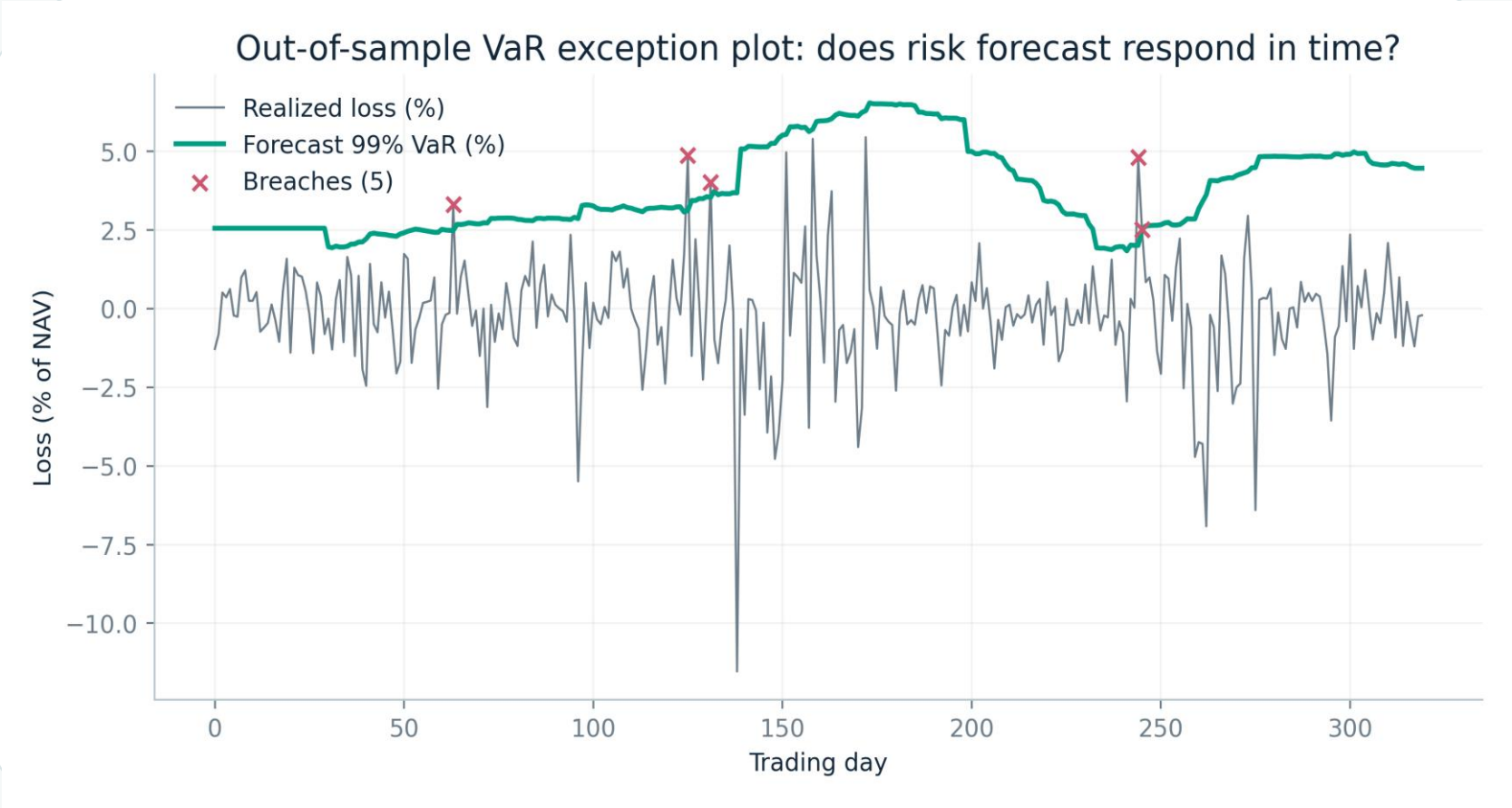
Stylized break-even analysis with equal-sized up/down moves and explicit costs.

ML success metrics versus investment success

Model metric	What it measures	What is missing
Accuracy / AUC	Classification separation	Trade costs and asymmetric payouts
RMSE / MAE	Numerical forecast error	Whether forecasts have usable signals
Sharpe / Sortino	Risk-adjusted realized strategy P&L	Skew, tails, implementation bias
Turnover / slippage	Implementability and friction	Unobserved liquidity under stress
Max drawdown / ES	Downside severity	Future unseen regimes / model uncertainty

Validate the entire trading strategy, not merely the forecast engine.

Backtesting: count breaches AND ask whether they cluster



READ THE CHART

- Kupiec: does the frequency match expected exceptions?
- Christoffersen: are exceptions independent over time?
- Clusters warn that regime shifts are being missed.

Illustrative one-day Gaussian VaR forecasts evaluated on held-out daily returns.

Avoid the four major research errors

1

Leakage

No future data in features or scaling

2

Selection

Do not tune thousands of rules on test data

3

Implementation

Apply realistic fees, rolls and rebalancing

4

Validation

Walk-forward and crisis/subsample tests

The backtest must reproduce information and constraints available at the trade date.

What the investment committee should receive

Panel	Question the decision maker asks	Evidence
Allocation	What exposures do you recommend?	Positions, cash and futures notional
Performance	Why is expected reward adequate?	Return assumptions and cost adjustment
Loss limit	What happens on a bad day?	VaR and ES in dollars
Tail robustness	What if losses are extreme?	EVT + copula / stress sensitivity
Model trust	Does the method work out of sample?	Exceptions, rolling validation
Governance	When will you exit or reduce risk?	Limits, rebalancing and caveats

Investment usefulness and reproducibility matter as much as mathematical sophistication.

AUDIENCE CHALLENGE

Final investment committee vote

USD 10m mandate. 99% daily VaR = $240k, ES = $360k, stress loss = $700k, futures margin calls can spike. Does a good expected return alone justify approval?

DEBRIEF No. Compare losses with stated risk tolerance, liquidity, suitability and model validation.

A credible recommendation can be “reduce overlay and reassess,” not always “buy.”

A competition-ready Streamlit research system

01 Market data: sources, adjustments, expiries, alignment and reproducible cleaning.

02 Positions: securities units, futures contracts, multipliers, collateral and roll calendar.

03 Research panels: return, drawdown, VaR / ES / EVT, model validation and stress scenarios.

04 Decision page: portfolio recommendation, cost assumptions, limits and weaknesses.

Ship one reproducible investment decision pipeline, not unrelated model outputs.

CLOSING

The investment researcher’s five commitments

Understand the asset. Trace the cash flows. Model the risks. Validate the forecasts. Defend the decision.

TRANSPARENCY

REPRODUCIBILITY

SUITABILITY

TECHNICAL APPENDIX

Formulas, conventions & methodological cautions

Reference material for deeper discussion and student research.

Return conventions and cash-flow consistency

$$R_t = \frac{P_t - P_{t-1} + D_t}{P_{t-1}} \quad ; \quad r_t = \ln(1 + R_t)$$

ECONOMIC INTERPRETATION

Simple returns are appropriate for NAV accounting and weighting across assets. Log returns add through time for a single positive-price wealth process; use cautiously for futures and multi-asset portfolios.

Note: treatment of distributions, dividends, interest, FX and split adjustments must be explicit.

Daily futures settlement and margin

$$VM_t = N \times M \times (F_t - F_{t-1})$$

ECONOMIC INTERPRETATION

Signed contracts N and multiplier M determine daily cash credited or debited to collateral. Margin deposit is performance security, not purchase cost.

Margin examples are illustrative, not current exchange requirements.

Cost of carry and commodity basis

$$F(0,T) \approx S_0 \exp[(r + u - y)T]$$

ECONOMIC INTERPRETATION

Futures pricing depends on financing, storage and convenience yield under simplifying no-arbitrage assumptions. Term structure differs by maturity, inventory tightness and delivery location.

Not a universal pricing identity for non-storable or constrained commodities.

Bond interest-rate sensitivity

$$\Delta P/P \approx -D_{mod} \times \Delta y + 0.5 \times \text{Convexity} \times (\Delta y)^2$$

ECONOMIC INTERPRETATION

Duration and convexity map a yield shock into approximate bond price P&L; credit spread moves require separate treatment.

Basis points: +100 bp = +0.01 decimal yield change.

Mean–variance portfolio optimizer

$$\min_{w} \left[\left(\frac{\lambda}{2} \right) w^{T} \Sigma w - w^{T} \mu \right] \; ; \; 1^{T} w = 1$$

ECONOMIC INTERPRETATION

The optimizer balances expected return and variance subject to budget, asset, margin and execution constraints. Weak estimates of μ can cause unstable extreme weights.

Add limits on turnover, futures gross notional, drawdown and cash reserve.

Marginal contribution to portfolio variance

$$\sigma_p = \sqrt{w^T \Sigma w} \quad ; \quad RC_i = \frac{w_i (\Sigma w)_i}{\sigma_p}$$

ECONOMIC INTERPRETATION

Risk contribution measures how much each allocation contributes to total volatility. Small weight can carry high risk if highly volatile or correlated.

Under differentiability; risk contribution to ES requires tail scenario distribution or suitable estimator.

Normal one-day VaR on portfolio returns

$$\mathrm{VaR}_{\alpha} \approx V \times [-\mu_r + z_{\alpha}\ \sigma_r]$$

ECONOMIC INTERPRETATION

A modeled high-quantile currency loss threshold, not the maximum loss. Assumes a normal loss model with correctly aligned horizon and return definition.

For a zero-mean return approximation, VaR_99% ≈ 2.326 σ × V.

Expected Shortfall for continuous losses

$$\mathrm{ES}_\alpha(L) = \frac{1}{1-\alpha}\int_a^1 \mathrm{VaR}_u(L)\,du$$

ECONOMIC INTERPRETATION

ES summarizes the average of the most adverse 1−α fraction of outcomes. It measures severity beyond the VaR cut-off and is coherent under standard definitions.

Sign conventions: positive L denotes portfolio loss.

EVT peaks over threshold model

$$G_{\xi,\beta}(y) = 1 - \left(1 + \frac{\xi y}{\beta}\right)^{-\frac{1}{\xi}}$$

ECONOMIC INTERPRETATION

For large losses exceeding u, fit a Generalized Pareto distribution to exceedances y=L−u. Tail index ξ diagnoses tail behaviour, but high-quantile uncertainty can be substantial.

Requires β>0 and 1+ξy/β>0. ξ→0 yields exponential limit.

Joint loss dependence and copulas

$$F_{X,Y}(x, y) = C\left[F_{X(x)}, F_{Y(y)}\right]$$

ECONOMIC INTERPRETATION

A copula separates marginal behaviour from dependence. Tail dependence is critical when several positions may incur extreme losses simultaneously.

Copula choice and uncertainty matter; Gaussian copula has zero asymptotic tail dependence when $|\rho|<1$.

Value-at-Risk exception testing

$$I_t = 1\{L_t > VaR_{\alpha,t}\}$$

ECONOMIC INTERPRETATION

Kupiec coverage compares breach frequency with 1−α. Independence tests examine clustering. Use truly out-of-sample daily forecasts and hold model selection outside the test set.

Traffic-light rules apply within their specified VaR confidence and backtest sample design.

Machine-learning signal economics

$$\mathrm{E}[\text{net P\&L}] = p \times \text{gain} - (1-p) \times \text{loss} - \text{all costs}$$

ECONOMIC INTERPRETATION

Classification probability by itself is not a trading decision. Include payoff asymmetry, frequency, turnover, execution, risk limits, calibration and tail exposure.

Purged/embargoed time-series CV is important when target windows overlap.

Copula lower-tail dependence coefficient

$$\lambda_L = \lim_{u \to 0+} P[\, U_2 \leq u \mid U_1 \leq u \,]$$

ECONOMIC INTERPRETATION

The coefficient measures the limiting probability that one asset falls into an extreme lower percentile when the other does. Gaussian and Student-t copulas can produce radically different joint crash behavior.

Tail dependence requires careful estimation: it is not equivalent to Pearson correlation.

Futures return on portfolio NAV

$$R_{p,t} \approx \frac{\Sigma_i \, n_i M_i \Delta F_{i,t} + \text{interest}_t - \text{costs}_t}{\text{NAV}_{t-1}}$$

ECONOMIC INTERPRETATION

Track signed contract counts n_i, price changes and multipliers, and explicitly add cash/collateral interest and trading costs. Using margin deposit as denominator answers a different leveraged-equity-return question.

Simplified self-financing overlay accounting; adapt for deposits/withdrawals, FX, taxes and portfolio rebalancing.

Wavelet coherence (conceptual estimator)

$$R^2(s,t) = \frac{CrossPower(s,t)}{Power_x(s,t) \cdot Power_y(s,t)}$$

ECONOMIC INTERPRETATION

This is a time–scale measure of co-movement after smoothing. A value near 1 indicates strong local coherence under the chosen wavelet and smoothing, not proof of predictive causality.

CrossPower = squared magnitude of smoothed cross-spectrum; Powerx, Powery = smoothed individual powers.

Selected research and practitioner sources

- CFTC — Economic Purpose of Futures Markets and How They Work (cftc.gov)
- CME Group — Introduction to Futures: Margin and Daily Settlement (cmegroup.com)
- CFA Institute — Introduction to Commodities and Commodity Derivatives (2026 curriculum)
- CFA Institute — Pricing and Valuation of Futures Contracts (2026 curriculum)
- Markowitz (1952); Engle (1982); Bollerslev (1986); Rockafellar & Uryasev (2000)
- McNeil, Frey & Embrechts — Quantitative Risk Management; Patton — Copula Methods
- Competition concept design (uploaded draft): mandate, technical layers and judging rubric